

A Centered-Extension and Defect-Minimization Strategy for Graph Reconstruction

Research sketch

May 3, 2026

Abstract

This note records an English proof strategy for the graph reconstruction conjecture. The main reframing is to fix one common card and view two reconstructions as one-vertex extensions of that common card. The problem then becomes: does the deck determine the attachment set of the missing vertex? The proposed proof uses a defect set between two attachment sets and tries to show that any nonzero defect can be repaired by moving the deleted “hole” to a better card alignment. The sharp local form of the problem is that a matched rooted card gives a color-preserving bijection whose graph error is concentrated in one star; the desired descent says that some such one-star error is smaller than the original color defect.

1 The centered-extension reduction

Let C be a finite simple graph with vertex set W . For a subset $A \subseteq W$, define $\text{Ext}(C, A)$ to be the graph obtained from C by adding one new vertex $*$, called the root, adjacent exactly to the vertices of A .

Thus

$$V(\text{Ext}(C, A)) = W \sqcup \{*\},$$

and

$$E(\text{Ext}(C, A)) = E(C) \cup \{\{*, u\} : u \in A\}.$$

For $u \in W$, deleting the old vertex u gives

$$\text{Ext}(C, A) - u \cong \text{Ext}(C - u, A \setminus \{u\}).$$

Deleting the root gives

$$\text{Ext}(C, A) - * \cong C.$$

Definition 1 (Punctured extension deck). *The punctured extension deck of the attachment set A over C is the multiset*

$$\text{PDeck}_C(A) = \{\{\text{Ext}(C - u, A \setminus \{u\}) : u \in W\}\},$$

where entries are taken up to graph isomorphism.

Then the full vertex deck of $\text{Ext}(C, A)$ is

$$\text{Deck}(\text{Ext}(C, A)) = \{\{C\}\} + \text{PDeck}_C(A).$$

Proposition 1 (Centered reduction). *Suppose G and H are graphs on $n \geq 3$ vertices with the same deck. Choose a matched card $G - x \cong H - y$, and identify both cards with a single graph C . Then there are subsets $A, B \subseteq V(C)$ such that*

$$G \cong \text{Ext}(C, A), \quad H \cong \text{Ext}(C, B),$$

and

$$\text{PDeck}_C(A) = \text{PDeck}_C(B).$$

Proof sketch. The subset A is the neighborhood of x , transported into the card $G - x \cong C$. Similarly B is the neighborhood of y , transported into $H - y \cong C$. Since the full decks of G and H agree and both contain the chosen matched card C , cancel one occurrence of C from the two deck multisets. $\square$

Thus the reconstruction conjecture follows from the following centered theorem.

Conjecture 1 (Centered extension theorem). *For every finite graph C and subsets $A, B \subseteq V(C)$, if*

$$\text{PDeck}_C(A) = \text{PDeck}_C(B),$$

then

$$\text{Ext}(C, A) \cong \text{Ext}(C, B).$$

Conjecture 2 (Root-preserving centered theorem). *If*

$$\text{PDeck}_C(A) = \text{PDeck}_C(B),$$

then A and B differ by an automorphism of C :

$$\exists \alpha \in \text{Aut}(C), \quad \alpha(A) = B.$$

Remark 1. *This root-preserving version implies the centered extension theorem. It is a stronger target because an isomorphism $\text{Ext}(C, A) \cong \text{Ext}(C, B)$ might in principle send the root of one extension to a non-root vertex of the other. Computationally, however, the stronger statement survives the small cases below. For now we treat it as the primary target, while retaining root-moving isomorphisms as a possible escape hatch in the moving-hole argument.*

Remark 2 (Computational evidence). *A brute-force search found no counterexample to the root-preserving centered theorem for all labeled cards C on at most five vertices, and no counterexample for the 156 unlabeled cards C on six vertices. The search buckets all attachment sets $A \subseteq V(C)$ by $\text{PDeck}_C(A)$ and checks that every bucket is a single $\text{Aut}(C)$ -orbit. This is only evidence, but it is useful: the stronger theorem survives the first small cases where nontrivial attachment-set buckets are abundant.*

Remark 3 (Computational evidence for descent). *In the same search range, a stronger one-step phenomenon held. Whenever $\text{PDeck}_C(A) = \text{PDeck}_C(B)$ and $A \neq B$, there was a defect vertex $u \in A \triangle B$, a vertex $v \in V(C)$, and a root-preserving isomorphism*

$$\text{Ext}(C - u, A \setminus \{u\}) \cong \text{Ext}(C - v, B \setminus \{v\})$$

whose recentering strictly reduced $|A \triangle B|$. For the 156 unlabeled six-vertex cards, this covered 11,008 nontrivial same-punctured-deck pairs. This suggests that the raw defect $|A \triangle B|$ may itself be the right induction parameter.

Remark 4 (Computational evidence for the single-switch base case). *A separate check of the single-switch situation*

$$A = S \cup \{p\}, \quad B = S \cup \{q\}$$

found no failure of the zero-error-match prediction for all unlabeled cards C on at most six vertices. Among the 156 unlabeled six-vertex cards, there were 37,440 single-switch pairs, 7,328 of which had equal punctured extension decks. Every same-punctured-deck pair had a root-preserving direct $p \leftrightarrow q$ match with zero star error. Exact two-error matches also occurred, in 1,240 of the same-punctured-deck pairs, but always alongside this direct zero-error companion. This is useful evidence for the corrected base-case target: exact two-error matches are allowed, but they should never be the only available defect-rooted matches. A follow-up check showed that detouring matches also occur frequently: among the six-vertex same-punctured-deck single switches, 3,000 admitted a root-preserving match from a defect vertex to a vertex of S . Thus detours are not obstructions by themselves; the evidence is instead that a direct zero-error representative is always also available. A classification of the same six-vertex data found that all 7,328 same-punctured-deck single switches satisfy $\deg_S(p) = \deg_S(q)$; 5,568 are settled by twin collapse, while 1,760 are non-twin and therefore represent the genuinely interesting balanced witness case.

2 Defect

Fix C and two attachment sets $A, B \subseteq V(C)$. Define the defect set

$$D(A, B) = A \Delta B.$$

Equivalently,

$$D_+ = A \setminus B, \quad D_- = B \setminus A, \quad D(A, B) = D_+ \sqcup D_-.$$

If $D(A, B) = \emptyset$, then the identity on C , extended by root to root, is an isomorphism

$$\text{Ext}(C, A) \cong \text{Ext}(C, B).$$

The plan is to prove that a nonempty defect cannot survive equality of punctured extension decks.

Conjecture 3 (No nontrivial centered trade). *If $\text{PDeck}_C(A) = \text{PDeck}_C(B)$, then either*

$$\text{Ext}(C, A) \cong \text{Ext}(C, B),$$

or there is another centered presentation of the same two extension graphs with strictly smaller defect.

Since defect size is finite, repeated repair would terminate at defect zero or at an isomorphism that moves the root. Either outcome proves reconstruction.

3 First shadow: edge counts

The defect is visible in aligned punctured shadows.

Lemma 1 (Aligned edge-count defect). *For each $u \in V(C)$,*

$$|E(\text{Ext}(C - u, A \setminus \{u\}))| - |E(\text{Ext}(C - u, B \setminus \{u\}))| = \mathbf{1}_{u \in B} - \mathbf{1}_{u \in A}.$$

Proof. Both graphs have the same base $C - u$. The first root has $|A| - \mathbf{1}_{u \in A}$ incident edges, while the second root has $|B| - \mathbf{1}_{u \in B}$ incident edges. Since equality of the full decks of $\text{Ext}(C, A)$ and $\text{Ext}(C, B)$ implies equality of their total edge counts, we have $|A| = |B|$. Subtracting gives the displayed formula. $\square$

Thus vertices in $A \setminus B$ and vertices in $B \setminus A$ create opposite edge-count errors in the aligned shadows. Equality of punctured decks can hide these errors only by rematching cards.

Remark 5. *This is the first conservation law. It cannot by itself prove the theorem: the deck is an unlabelled multiset, so the card indexed by u on the A -side may be matched to a card indexed by some different v on the B -side. The proof must exploit the entire isomorphism type of each punctured extension, not just its number of edges.*

4 Higher shadows

Let F be any finite graph. Consider the difference

$$s_F(\text{Ext}(C - u, A \setminus \{u\})) - s_F(\text{Ext}(C - u, B \setminus \{u\})),$$

where $s_F(X)$ denotes the number of induced copies of F in X . Copies of F that avoid the root depend only on $C - u$ and cancel. The entire difference is caused by copies of F using the root. Hence it is a function of the rooted neighborhood data $A \setminus \{u\}$ versus $B \setminus \{u\}$ inside $C - u$.

This gives a family of constraints:

$$\begin{aligned} & \{ \{ \text{rooted pattern profile of } (C - u, A \setminus \{u\}) : u \in V(C) \} \} \\ &= \{ \{ \text{rooted pattern profile of } (C - u, B \setminus \{u\}) : u \in V(C) \} \}. \end{aligned}$$

Equivalently, for every finite pattern F , let $r_F(C - u, A \setminus \{u\})$ be the number of induced copies of F in $\text{Ext}(C - u, A \setminus \{u\})$ that use the root. Then

$$\{ \{ r_F(C - u, A \setminus \{u\}) : u \in V(C) \} \} = \{ \{ r_F(C - u, B \setminus \{u\}) : u \in V(C) \} \}.$$

The centered theorem says that no non-isomorphic pair of attachment sets can satisfy all of these constraints simultaneously.

5 The moving-hole repair lemma

The most promising proof mechanism is to use an isomorphism between two matched punctured shadows to move the deleted vertex, or “hole”.

Suppose

$$\theta : \text{Ext}(C - u, A \setminus \{u\}) \cong \text{Ext}(C - v, B \setminus \{v\})$$

is one of the card isomorphisms supplied by $\text{PDeck}_C(A) = \text{PDeck}_C(B)$.

Recenter both full extensions at the deleted old vertices u and v . The new common card is the left punctured extension

$$D = \text{Ext}(C - u, A \setminus \{u\}),$$

and the right punctured extension is identified with D by θ^{-1} .

In the left full graph $\text{Ext}(C, A)$, the missing vertex u is adjacent in the new card D to

$$A_u^{\text{new}} = N_C(u) \setminus \{u\} \cup \begin{cases} \{*_A\}, & u \in A, \\ \emptyset, & u \notin A, \end{cases}$$

where $*_A$ is the root of the left extension. In the right full graph, the missing vertex v is adjacent in its punctured card to

$$B_v^{\text{raw}} = N_C(v) \setminus \{v\} \cup \begin{cases} \{*_B\}, & v \in B, \\ \emptyset, & v \notin B. \end{cases}$$

After transporting to the common card D , the right attachment set is

$$B_v^{\text{new}} = \theta^{-1}(B_v^{\text{raw}}).$$

Thus the recentered defect is

$$\Delta_\theta(u, v) = A_u^{\text{new}} \Delta B_v^{\text{new}}.$$

The desired repair statement is therefore concrete:

$$|\Delta_\theta(u, v)| < |A \Delta B|$$

unless the recentering already exhibits an isomorphism of the full extensions.

There are two cases.

5.1 Case 1: the root maps to the root

Then θ restricts to an isomorphism

$$\psi : C - u \cong C - v$$

and carries $A \setminus \{u\}$ to $B \setminus \{v\}$. The new defect formula simplifies to

$$\Delta_\theta(u, v) = (N_C(u) \setminus \{u\}) \Delta \psi^{-1}(N_C(v) \setminus \{v\}).$$

Indeed, because θ maps root to root, it preserves root degree:

$$|A| - \mathbf{1}_{u \in A} = |B| - \mathbf{1}_{v \in B}.$$

Since the full extensions have equal edge count, $|A| = |B|$, and hence $\mathbf{1}_{u \in A} = \mathbf{1}_{v \in B}$. Thus the possible root term cancels.

For an arbitrary unrooted card isomorphism, without assuming root maps to root, we only get the weaker conservation law

$$\deg_C(u) + \mathbf{1}_{u \in A} = \deg_C(v) + \mathbf{1}_{v \in B}.$$

Equivalently, if root membership changes in an unrooted match, it is compensated by a one-unit change in the old-card degree.

Lemma 2 (Empty recentered defect extends). *In the root-preserving case, if*

$$\Delta_\theta(u, v) = \emptyset,$$

then

$$\text{Ext}(C, A) \cong \text{Ext}(C, B).$$

Proof sketch. The equality $\Delta_\theta(u, v) = \emptyset$ says that ψ carries the neighborhood of u in C to the neighborhood of v in C . Therefore ψ extends over $u \mapsto v$ to an isomorphism $C \cong C$. The root-preserving card isomorphism also gives $\psi(A \setminus \{u\}) = B \setminus \{v\}$, and root-degree preservation gives $u \in A$ if and only if $v \in B$. Hence the extended isomorphism carries A to B , and then extends root-to-root to an isomorphism of $\text{Ext}(C, A)$ and $\text{Ext}(C, B)$. $\square$

5.2 Star-error reformulation

The root-preserving case can be stated more cleanly without mentioning the new centered graph D . Given a root-preserving card isomorphism

$$\theta : \text{Ext}(C - u, A \setminus \{u\}) \cong \text{Ext}(C - v, B \setminus \{v\}),$$

let

$$\psi : C - u \cong C - v$$

be its restriction to old vertices, and extend ψ to a bijection

$$\hat{\psi} : V(C) \rightarrow V(C)$$

by setting $\hat{\psi}(u) = v$.

Lemma 3 (A rooted card match is a one-star graph error). *In the root-preserving case, $\hat{\psi}(A) = B$. Moreover $\hat{\psi}$ preserves every edge and non-edge of C except possibly pairs incident to u . Its failure to be an automorphism is exactly*

$$\text{Err}_u(\hat{\psi}) = \{ w \in V(C) \setminus \{u\} : (uw \in E(C)) \not\leftrightarrow (\hat{\psi}(u)\hat{\psi}(w) \in E(C)) \}.$$

Under the recentring identification,

$$|\text{Err}_u(\hat{\psi})| = |\Delta_\theta(u, v)|.$$

Proof sketch. The card isomorphism gives $\psi(A \setminus \{u\}) = B \setminus \{v\}$. Root degree preservation, together with $|A| = |B|$, gives $u \in A$ if and only if $v \in B$. Hence the extension $\hat{\psi}$ carries all of A to all of B . Since ψ is an isomorphism $C - u \cong C - v$, every adjacency relation not involving u is preserved. The only possible failures are the adjacencies from u , and the set of such failures is precisely the neighborhood symmetric difference appearing in $\Delta_\theta(u, v)$. $\square$

Lemma 4 (Parity of star error). *In the same root-preserving situation,*

$$\deg_C(u) = \deg_C(v),$$

and therefore

$$|\text{Err}_u(\hat{\psi})| = |(N_C(u) \setminus \{u\}) \Delta \psi^{-1}(N_C(v) \setminus \{v\})|$$

is even.

Proof sketch. The restricted map $\psi : C - u \cong C - v$ gives

$$|E(C - u)| = |E(C - v)|.$$

Since

$$|E(C - u)| = |E(C)| - \deg_C(u) \quad \text{and} \quad |E(C - v)| = |E(C)| - \deg_C(v),$$

the deleted vertices have equal degree. The two neighborhoods compared in the symmetric difference therefore have equal cardinality, so their symmetric difference has even size. $\square$

The parity lemma exposes a genuine base case. If

$$|A \Delta B| = 2,$$

then a strict descent move must have star error 0, not merely a smaller positive error. Thus the first nontrivial local theorem is the following.

Conjecture 4 (Single-switch rigidity). *Suppose*

$$A = S \cup \{p\}, \quad B = S \cup \{q\}, \quad p \neq q,$$

with $p, q \notin S$, and suppose

$$\text{PDeck}_C(A) = \text{PDeck}_C(B).$$

Then $\text{Ext}(C, A) \cong \text{Ext}(C, B)$. In the root-preserving version, there is an automorphism of C carrying the set $S \cup \{p\}$ to the set $S \cup \{q\}$.

This should be treated as the base case for the whole descent strategy. Computationally it is always resolved by a zero-error card match in the tested range. Conceptually, it says that the deck cannot hide the movement of a single colored vertex unless the card already has the corresponding symmetry.

5.3 Single-switch constraints

For the rest of the single-switch analysis, work in the root-preserving regime: either root-moving matches have already given success/ambiguity descent, or we have extracted a root-preserving trade. The rooted card-match graph and all root-degree arguments below are statements in that regime.

In the single-switch situation, the two exceptional aligned shadows are visible even before using the full isomorphism type of the cards. For

$$A = S \cup \{p\}, \quad B = S \cup \{q\},$$

the aligned edge-count defect is

$$|E(\text{Ext}(C - p, S))| - |E(\text{Ext}(C - p, S \cup \{q\}))| = -1$$

and

$$|E(\text{Ext}(C - q, S \cup \{p\}))| - |E(\text{Ext}(C - q, S))| = +1.$$

All other aligned shadows have edge-count defect 0. Therefore any deck matching must compensate for the missing root edge at p by rematching cards; it cannot simply align every old vertex with itself.

Lemma 5 (Single-switch degree balance). *If $\text{PDeck}_C(S \cup \{p\}) = \text{PDeck}_C(S \cup \{q\})$, then*

$$\deg_C(p) = \deg_C(q).$$

Proof sketch. The full extensions have the same vertex deck, so they have the same degree multiset. The root has the same degree on both sides, namely $|S| + 1$. Every old vertex except p and q has the same full-extension degree on both sides. At p and q , the two multisets are

$$\{\{\deg_C(p) + 1, \deg_C(q)\}\} \quad \text{and} \quad \{\{\deg_C(p), \deg_C(q) + 1\}\}.$$

Equality of these two multisets forces $\deg_C(p) = \deg_C(q)$. □

Lemma 6 (Single-switch S -degree balance). *If $\text{PDeck}_C(S \cup \{p\}) = \text{PDeck}_C(S \cup \{q\})$, then*

$$\deg_S(p) = \deg_S(q).$$

Proof sketch. The full decks of

$$\text{Ext}(C, S \cup \{p\}) \quad \text{and} \quad \text{Ext}(C, S \cup \{q\})$$

are equal, because both are obtained from the common root-deleted card C plus the equal punctured extension decks. Therefore they have the same number of triangles, since triangle counts are reconstructible from the vertex deck by Kelly's lemma.

Triangles avoiding the root are just triangles of C , hence cancel. The root-using triangles are exactly the edges induced by the attachment set. On the A -side this number is

$$|E(C[S])| + \deg_S(p),$$

and on the B -side it is

$$|E(C[S])| + \deg_S(q).$$

Equality of triangle counts gives $\deg_S(p) = \deg_S(q)$. $\square$

Lemma 7 (Root-link profile balance). *In the same single-switch setting, let F be any finite graph with fewer vertices than $\text{Ext}(C, S \cup \{p\})$. Then the number of induced copies of F in $\text{Ext}(C, S \cup \{p\})$ that use the added root is equal to the number of induced copies of F in $\text{Ext}(C, S \cup \{q\})$ that use the added root.*

Proof sketch. The full decks of the two extensions are equal, so Kelly's lemma gives equal total induced F -counts. Copies of F that avoid the added root lie entirely in the common card C , so their counts are identical on the two sides. Subtracting these common root-avoiding counts gives equality of the root-using counts. $\square$

This is the real source of the low-order balances. Taking $F = K_3$ gives $\deg_S(p) = \deg_S(q)$. Larger choices of F say that the way p interacts with S , and more generally with all of C , has the same root-visible profile as the way q interacts with S . A non-twin single-switch state must therefore hide its witnesses from every proper root-link count; the only remaining distinction is how those witnesses are arranged globally in the card matching.

Now suppose the p -shadow on the A -side is matched root-preservingly to the v -shadow on the B -side:

$$\text{Ext}(C - p, S) \cong \text{Ext}(C - v, B \setminus \{v\}).$$

Root degree preservation forces

$$|S| = |B| - \mathbf{1}_{v \in B} = |S| + 1 - \mathbf{1}_{v \in B},$$

so $v \in B$. Thus the A -side deleted defect vertex p can only be matched to a B -colored vertex $v \in S \cup \{q\}$. Symmetrically, the B -side deleted defect vertex q can only be matched to an A -colored vertex.

This is a useful first step: in defect 2, the trade must move deleted defect vertices into the colored set on the other side. If one of these matches has zero star error, we are done. If not, parity says its star error is at least 2, which is exactly the original defect size. Hence a minimal single-switch counterexample would have the following rigid form: every defect-rooted root-preserving card match has star error exactly 2 or larger, and no zero-error match exists.

Conjecture 5 (No exact two-error single switch). *The rigid form just described cannot occur. More concretely, if*

$$A = S \cup \{p\}, \quad B = S \cup \{q\},$$

and $\text{PDeck}_C(A) = \text{PDeck}_C(B)$, then some root-preserving card match based at p or q has zero star error.

Conjecture 6 (Direct single-switch card match). *In the same single-switch situation, deck equality already forces a root-preserving isomorphism*

$$\text{Ext}(C - p, S) \cong \text{Ext}(C - q, S)$$

whose recentered star error is zero. Equivalently, there is an isomorphism

$$\psi : C - p \cong C - q$$

with $\psi(S) = S$ that extends by $p \mapsto q$ to an automorphism of C .

This is stronger than merely asking for some zero-error defect-rooted match. It says that the two natural defect-deleted shadows already match each other. Computationally this stronger form holds in the tested range. If true, it proves the single-switch base case in one shot.

5.4 Detours in the single-switch trade cycle

The direct card-match conjecture isolates the only way the base case can fail. The natural p -deleted shadow

$$\text{Ext}(C - p, S)$$

has root degree $|S|$. On the B -side, a root-preserving match with the same root degree must delete a vertex $v \in B = S \cup \{q\}$. If $v = q$, we are in the direct case. Thus a failure of the direct conjecture means that every trade sends p first to some vertex of S , creating a detour around the unique defect cycle before it reaches q .

Choose a root-preserving trade and let

$$p = x_0, x_1, \dots, x_k = q$$

be the directed arc of the τ -cycle from p to q , where $\tau(x_i) = x_{i+1}$. In the non-direct case, $k \geq 2$, and the internal vertices $x_1, \dots, x_{k-1}$ lie in S . Along this arc, the rooted cards give isomorphisms

$$\text{Ext}(C - x_i, A \setminus \{x_i\}) \cong \text{Ext}(C - x_{i+1}, B \setminus \{x_{i+1}\}).$$

For $1 \leq i < k$, both x_i and x_{i+1} lie in S , so the color sets differ by moving the hole along S while keeping the switched vertex on opposite sides. The final step into q is the only step that can repair the original missing color.

Conjecture 7 (Single-switch detour shortening). *In a minimal single-switch counterexample, one can choose a root-preserving trade with no detour arc*

$$p = x_0 \rightarrow x_1 \rightarrow \dots \rightarrow x_k = q$$

with $k \geq 2$. Equivalently, one can choose a trade with $\tau(p) = q$.

This is a plausible intermediate target between raw deck equality and the direct card-match conjecture. The rooted invariant cycle-balance laws apply on the whole p -to- q cycle, while the internal detour vertices have zero edge-count charge. If an internal detour exists, one hopes to splice the cycle or recenter at an internal vertex to obtain either a shorter detour or a zero-error direct match.

Detouring trades are not themselves pathological. They occur frequently in small terminal examples, sometimes even with zero star error. The point is selection, not exclusion: the deck may admit many trades, but the conjecture asks for at least one direct representative.

In the rooted card-match graph, this says that the edge $p \rightsquigarrow q$ should exist. If a particular perfect matching uses a detour edge $p \rightsquigarrow s$ with $s \in S$, that is not a contradiction; it only means the matching has chosen a non-direct representative. The selection problem is to show that the match graph contains a direct good edge, and preferably that Hall alternation can modify any detouring perfect matching to one using it.

Lemma 8 (Detour arcs are degree-flat). *Let*

$$p = x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_k = q$$

be the p -to- q arc of a root-preserving single-switch trade, with internal vertices in S . Then

$$\deg_C(x_0) = \deg_C(x_1) = \cdots = \deg_C(x_k).$$

Proof sketch. For any trade edge $x_i \rightsquigarrow x_{i+1}$, equality of the edge counts of the matched rooted cards gives

$$\deg_C(x_i) + \mathbf{1}_{x_i \in A} = \deg_C(x_{i+1}) + \mathbf{1}_{x_{i+1} \in B}.$$

Along the p -to- q arc, $x_i \in A$ and $x_{i+1} \in B$ at every step: $p \in A$, $q \in B$, and every internal detour vertex lies in $S \subseteq A \cap B$. The indicator terms therefore cancel at each step. $\square$

More generally, every rooted invariant gives a transport equation along the detour:

$$I_A(x_i) = I_B(x_{i+1}).$$

For $I = T$ and $I = P$, these equations relate the rooted triangle and two-step profiles of consecutive detour vertices. A detour is therefore a string of vertices that are not merely degree-equal; they are successively indistinguishable by the rooted card profiles determined by the switch.

Lemma 9 (Natural shadow isolation criterion). *In the single-switch setting, suppose the rooted card*

$$\text{Ext}(C - p, S)$$

is not isomorphic to any card of the form

$$\text{Ext}(C - s, (S \setminus \{s\}) \cup \{q\}) \quad (s \in S).$$

Then

$$\text{Ext}(C - p, S) \cong \text{Ext}(C - q, S).$$

Proof sketch. The card $\text{Ext}(C - p, S)$ appears on the A -side punctured extension deck. By equality of punctured decks it must appear on the B -side. Root degree shows that any root-preserving occurrence on the B -side must delete a vertex of $B = S \cup \{q\}$. By hypothesis it cannot be obtained by deleting a vertex of S , so it must be the q -deleted card. $\square$

The difficult case is therefore exactly the non-isolated case: the natural p -shadow is profile-equivalent to at least one detour shadow. The degree-flat lemma is the first invariant obstruction to such absorption; the rooted triangle and rooted two-step transport equations provide stronger filters.

5.5 Shadow-type flow

There is a useful multiplicity refinement of the isolation criterion. Let

$$T_p = \text{Ext}(C - p, S)$$

be the natural p -deleted rooted card type. Define

$$L(T_p) = \{u \in V(C) : \text{Ext}(C - u, A \setminus \{u\}) \cong T_p\},$$

and

$$R(T_p) = \{v \in V(C) : \text{Ext}(C - v, B \setminus \{v\}) \cong T_p\}.$$

Equality of punctured decks gives

$$|L(T_p)| = |R(T_p)|.$$

The vertex p always lies in $L(T_p)$. The vertex q lies in $R(T_p)$ exactly when the direct edge $p \rightsquigarrow q$ exists. The vertices of $R(T_p) \cap S$ are precisely the p -absorbers.

Thus if the direct edge is absent, the p -mass in the deck must be absorbed inside S :

$$q \notin R(T_p) \implies R(T_p) \subseteq S.$$

Multiplicity balance then forces

$$|L(T_p) \setminus \{p\}| = |R(T_p)|.$$

So every failed direct edge produces not only absorbers on the B -side, but also additional suppliers on the A -side: vertices

$$u \neq p \quad \text{with} \quad \text{Ext}(C - u, A \setminus \{u\}) \cong \text{Ext}(C - p, S).$$

Conjecture 8 (No closed shadow-type flow). *In the single-switch setting, the natural shadow type T_p cannot be supported entirely by a closed supplier-absorber flow avoiding q . In other words, multiplicity balance for T_p , together with the absorber constraints for every vertex of $R(T_p) \cap S$, forces $q \in R(T_p)$.*

This is a sharper form of detour selection. Instead of following one chosen perfect matching, it examines the whole fiber of the natural card type. A counterexample would need a balanced population of T_p -suppliers and T_p -absorbers, all satisfying the same rooted profile constraints, with the natural destination q missing from the right fiber.

There is also a dual flow. Let

$$T_q = \text{Ext}(C - q, S)$$

be the natural q -deleted rooted card type on the B -side. If the direct edge $p \rightsquigarrow q$ is absent, then $T_p \neq T_q$. The type T_q appears at q on the B -side, and equality of punctured decks forces it to appear on the A -side. Root degree again shows that every such A -side occurrence must delete a vertex of $A = S \cup \{p\}$. Since deleting p would give T_p , not T_q , the T_q -mass must also be routed through S .

Conjecture 9 (No coupled closed shadow flows). *In a single-switch same-deck state, it is impossible for both natural shadow types*

$$T_p = \text{Ext}(C - p, S), \quad T_q = \text{Ext}(C - q, S)$$

to be routed entirely through the common set S , avoiding the direct positions q and p , respectively.

This coupled-flow conjecture is another form of direct-edge selection. It says that the common set S cannot simultaneously absorb all evidence of the p -deleted natural card and all evidence of the q -deleted natural card. The two flows would have to pass through vertices with the same total degree, the same S -degree constraints, and compatible rooted profiles. The expectation is that these constraints force the two natural shadow types to coincide after all, producing the direct edge.

Definition 2 (Supplier). *In the same setting, a vertex $u \in S$ is a p -supplier if*

$$\text{Ext}(C - u, (S \setminus \{u\}) \cup \{p\}) \cong \text{Ext}(C - p, S).$$

Equivalently, $u \in L(T_p) \setminus \{p\}$.

Lemma 10 (Supplier constraints). *If $u \in S$ is a p -supplier, then*

$$\deg_C(u) = \deg_C(p)$$

and

$$\deg_S(u) = \deg_{S \setminus \{u\}}(p).$$

Proof sketch. The root degree of the u -deleted A -side card is

$$|A| - \mathbf{1}_{u \in A}.$$

To match $T_p = \text{Ext}(C - p, S)$, whose root degree is $|S|$, we must have $u \in A$. Since $u \neq p$, this puts $u \in S$. Equality of total edge counts then gives $\deg_C(u) = \deg_C(p)$.

For the second equality, compare root-using triangle counts. The natural p -shadow has $|E(C[S])|$ such triangles. The u -deleted A -shadow has color set $(S \setminus \{u\}) \cup \{p\}$, so its root-using triangle count is

$$|E(C[S])| - \deg_S(u) + \deg_{S \setminus \{u\}}(p).$$

Equality gives the displayed constraint. □

Definition 3 (Absorber). *In the single-switch setting, a vertex $s \in S$ is a p -absorber if there is a root-preserving isomorphism*

$$\text{Ext}(C - p, S) \cong \text{Ext}(C - s, (S \setminus \{s\}) \cup \{q\}).$$

Equivalently, the natural p -deleted shadow can be matched to the s -deleted shadow on the B -side.

Lemma 11 (Absorber constraints). *Suppose $s \in S$ is a p -absorber. Then the following numerical constraints hold:*

$$\deg_C(s) = \deg_C(p),$$

$$\deg_S(s) = \deg_{S \setminus \{s\}}(q),$$

and, assuming the single-switch degree balance $\deg_C(p) = \deg_C(q)$,

$$\deg_S(p) = \deg_S(q).$$

Proof sketch. The first equality comes from equality of the total edge counts of the matched rooted cards:

$$|E(C - p)| + |S| = |E(C - s)| + |S|.$$

For the second equality, compare root-using triangle counts. The p -shadow has

$$|E(C[S])|$$

root-using triangles, while the s -deleted B -shadow has

$$|E(C[(S \setminus \{s\}) \cup \{q\}])| = |E(C[S])| - \deg_S(s) + \deg_{S \setminus \{s\}}(q).$$

Equality gives the displayed triangle constraint.

For the last equality, compare the rooted two-step counts P . The p -shadow gives

$$\sum_{x \in S} \deg_{C-p}(x) = \sum_{x \in S} \deg_C(x) - \deg_S(p).$$

The s -deleted B -shadow gives

$$\sum_{x \in S \setminus \{s\}} \deg_{C-s}(x) + \deg_{C-s}(q) = \sum_{x \in S} \deg_C(x) - \deg_C(s) - \deg_S(s) + \deg_C(q) - \mathbf{1}_{sq \in E(C)}.$$

Using $\deg_C(s) = \deg_C(p) = \deg_C(q)$, this becomes

$$\sum_{x \in S} \deg_C(x) - \deg_S(s) - \mathbf{1}_{sq \in E(C)}.$$

Together with

$$\deg_S(s) = \deg_{S \setminus \{s\}}(q) = \deg_S(q) - \mathbf{1}_{sq \in E(C)},$$

the two-step equality gives $\deg_S(p) = \deg_S(q)$. □

Thus absorption is already quite restrictive. A p -absorber must have the same total degree as p , must see S exactly as q sees $S \setminus \{s\}$, and is compatible with the global S -degree balance forced by triangle reconstruction. If these absorber constraints fail for every $s \in S$, the natural p -shadow is isolated and the direct $p \rightsquigarrow q$ edge is forced.

Therefore the hard detour-selection case is not just degree-balanced in C ; it is also balanced relative to the common color:

$$\deg_C(p) = \deg_C(q), \quad \deg_S(p) = \deg_S(q).$$

Equivalently, p and q have the same number of neighbors in $V(C) \setminus (S \cup \{p, q\})$ as well. The remaining witnesses must be balanced both inside and outside the common color.

5.6 Absorber chains

Absorbers should be organized into paths, not treated as isolated accidents. Define the absorption digraph for the single-switch state as follows. Its vertices are

$$S \cup \{q\}.$$

There is an edge

$$p \rightarrow y$$

if the natural p -deleted shadow can be matched root-preservingly to the y -deleted B -side shadow. Thus $y = q$ is the desired direct edge, and $y \in S$ is a p -absorber. More generally, along a chosen root-preserving trade cycle

$$p = x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_k = q,$$

each step $x_i \rightarrow x_{i+1}$ is an absorption edge for the current deleted shadow.

For any such step, equality of rooted card profiles gives transport equations

$$I_A(x_i) = I_B(x_{i+1})$$

for every rooted invariant I . In particular, the whole chain is degree-flat:

$$\deg_C(x_0) = \deg_C(x_1) = \cdots = \deg_C(x_k),$$

and it is also flat for every invariant that can be read from the rooted card after correcting for the moving color set.

Conjecture 10 (Absorber-chain shortening). *If there is an absorption chain*

$$p = x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_k = q$$

with $k \geq 2$, then either there is a direct edge $p \rightarrow q$, or there is a shorter absorption chain from p to q .

This is the Hall-alternation version of detour shortening. The hope is to splice a chain at an internal absorber using the fact that all vertices on the chain have compatible rooted profiles. The absorber constraints above are the first low-order consequences of this compatibility; higher rooted subgraph counts should make long chains increasingly rigid.

5.7 Choosing the extendable card isomorphism

Even after detour shortening, there is one remaining issue. A direct root-preserving match gives

$$\psi : C - p \cong C - q, \quad \psi(S) = S.$$

This proves that p and q are deletion-similar relative to the color set S , but it does not by itself prove that ψ extends over $p \mapsto q$. Extension is exactly the zero-star-error condition

$$\psi(N_C(p) \setminus \{p\}) = N_C(q) \setminus \{q\}.$$

Thus the direct single-switch theorem has two layers:

$$\text{find a direct card match } C - p \cong C - q,$$

then

$$\text{choose such a match that also carries } N_C(p) \text{ to } N_C(q).$$

The first layer is detour shortening. The second layer is neighborhood selection inside the isomorphism set

$$\text{Iso}_S(C - p, C - q).$$

Conjecture 11 (Neighborhood selection). *Assume $\text{PDeck}_C(S \cup \{p\}) = \text{PDeck}_C(S \cup \{q\})$, and assume $\text{Iso}_S(C - p, C - q)$ is nonempty. Then there exists*

$$\psi \in \text{Iso}_S(C - p, C - q)$$

such that

$$\psi(N_C(p) \setminus \{p\}) = N_C(q) \setminus \{q\}.$$

This is a colored version of the familiar obstruction that two vertices may have isomorphic deleted cards without being related by an automorphism. The extra hypothesis here is not merely $C - p \cong C - q$; it is equality of the entire punctured extension decks for the two one-vertex colorings. The conjecture says that this extra deck information selects an extendable isomorphism of the common deleted card.

5.8 Stabilizer-orbit formulation

Neighborhood selection can be made completely algebraic. Fix one direct card isomorphism

$$\psi_0 : C - p \cong C - q, \quad \psi_0(S) = S.$$

Let

$$K = C - p, \quad U = N_C(p) \setminus \{p\}, \quad V_0 = \psi_0^{-1}(N_C(q) \setminus \{q\}).$$

Let

$$\text{Aut}(K, S) = \{\alpha \in \text{Aut}(K) : \alpha(S) = S\}.$$

Lemma 12 (Neighborhood selection is an orbit problem). *There exists an S -preserving isomorphism*

$$\psi : C - p \cong C - q$$

that extends over $p \mapsto q$ if and only if U and V_0 are in the same $\text{Aut}(K, S)$ -orbit.

Proof sketch. Every other S -preserving isomorphism $C - p \cong C - q$ has the form

$$\psi = \psi_0 \alpha$$

with $\alpha \in \text{Aut}(K, S)$. The transported q -neighborhood is then

$$\psi^{-1}(N_C(q) \setminus \{q\}) = \alpha^{-1}(V_0).$$

Thus ψ has zero star error exactly when

$$U = \alpha^{-1}(V_0),$$

equivalently when U and V_0 lie in the same orbit. □

So a failure of neighborhood selection is not vague. It is a pair of subsets

$$U, V_0 \subseteq V(K)$$

that are indistinguishable enough to arise from the same full punctured-extension deck, but still lie in distinct $\text{Aut}(K, S)$ -orbits. This is the point where group action can replace some of the deck language.

Lemma 13 (Finite orbit separator). *If U and V_0 are not in the same $\text{Aut}(K, S)$ -orbit, then there is a finite two-colored rooted pattern F whose number of induced occurrences in the colored structure*

$$(K, S, U)$$

differs from its number of induced occurrences in

$$(K, S, V_0).$$

Proof sketch. The two finite colored structures (K, S, U) and (K, S, V_0) are non-isomorphic exactly when U and V_0 are in distinct $\text{Aut}(K, S)$ -orbits. Taking F to be the whole colored structure (K, S, U) already separates them: it occurs in itself, with multiplicity $|\text{Aut}(K, S, U)|$, and has no surjective color-preserving induced occurrence in the non-isomorphic structure. One can then choose a separator of minimal size or complexity. $\square$

The minimal separator F is a possible measure of neighborhood-selection failure. If F is small, it should appear in rooted triangle, path, or other low-order shadow counts. If F is large, then all smaller colored patterns agree, which says U and V_0 are locally indistinguishable relative to S . Either outcome is useful: a small separator feeds cycle balance, while a large minimal separator gives a strong local-equivalence normal form.

Lemma 14 (Anchored Kelly pressure). *Suppose the distinguished root information is available in the deck, so that we can compare the one-vertex-deleted cards as rooted structures over the same first root. Then equality of the anchored decks of*

$$(K, S, U) \quad \text{and} \quad (K, S, V)$$

implies equality of the number of induced occurrences of every proper two-colored pattern F , meaning every F with fewer vertices than K .

Proof sketch. This is the colored version of Kelly's lemma. Count pairs

$$(W, z)$$

where W is an induced occurrence of F and z is a vertex outside that occurrence. Each occurrence of F is counted $|V(K)| - |V(F)|$ times, and the same pair count can be read from the anchored deck by summing the number of F -occurrences in each one-vertex-deleted anchored card. $\square$

Consequently, under anchored deck equality, a finite orbit separator for (K, S, U) and (K, S, V) cannot be proper. Any failure of neighborhood selection must be a full-size colored hypomorphism: all proper colored patterns agree, but the full colored structures are not isomorphic. This is exactly the kind of rigid boundary case where reconstruction difficulties usually live.

The caveat is important. The original deck is unrooted. The anchored Kelly pressure applies only after the root-moving ambiguity has been controlled, or inside a strengthened anchored subproblem where the distinguished root is part of the data. Still, it explains why low-order invariants either quickly prove neighborhood selection or force the bad case into a very narrow full-size hypomorphic corner.

5.9 Fixed-host subset reconstruction

The neighborhood-selection problem can be isolated from graph reconstruction as follows. Fix a colored host

$$(K, S).$$

For a subset $U \subseteq V(K)$, define its fixed-host deletion deck

$$\text{Deck}_{K,S}(U) = \{ \{ (K - z, S \setminus \{z\}, U \setminus \{z\}) : z \in V(K) \} \},$$

where the cards are taken up to color-preserving graph isomorphism.

Conjecture 12 (Fixed-host subset reconstruction). *If*

$$\text{Deck}_{K,S}(U) = \text{Deck}_{K,S}(V),$$

and

$$|U| = |V|,$$

then U and V are in the same $\text{Aut}(K, S)$ -orbit.

This conjecture is exactly the anchored neighborhood-selection problem once the first root has been recognized. It is weaker-looking than graph reconstruction because the uncolored host K and the first color S are held fixed. The only unknown is the second color class. The equal-size hypothesis is essential: for a one-vertex host, the deletion deck cannot distinguish the empty second color from the singleton second color after the only vertex is deleted. In the graph-reconstruction application, equal size is supplied by root-degree equality.

Remark 6. *The fixed-host problem is still nontrivial. The cards are not labelled subsets of $K - z$; they are isomorphism types of colored deleted hosts. A host with many deletion-similar vertices can hide the location of U , which is exactly the phenomenon behind absorbers and detours. But the problem has a finite group-action target: prove orbit equality under $\text{Aut}(K, S)$, not construct an arbitrary graph isomorphism.*

Anchored Kelly pressure says that any counterexample to fixed-host subset reconstruction must have all proper colored pattern counts equal. Thus a minimal counterexample is a fixed-host colored hypomorphism:

$$(K, S, U) \quad \text{and} \quad (K, S, V)$$

have equal colored deletion decks and equal proper colored substructure counts, but U and V are in different $\text{Aut}(K, S)$ -orbits.

Remark 7 (Computational evidence). *A brute-force check found no counterexample to equal-size fixed-host subset reconstruction for all unlabeled hosts K on at most five vertices, all choices of S , and all equal-size pairs U, V . The same check without the equal-size hypothesis fails immediately on the one-vertex host, as noted above.*

5.10 Fixed-host trades and one-star errors

The fixed-host obstruction has the same local shape as the centered obstruction. Assume

$$\text{Deck}_{K,S}(U) = \text{Deck}_{K,S}(V), \quad |U| = |V|.$$

Choose a matching of the fixed-host decks. Thus for each $z \in V(K)$ there is a vertex $\tau(z) \in V(K)$ and a color-preserving isomorphism

$$\eta_z : (K - z, S \setminus \{z\}, U \setminus \{z\}) \cong (K - \tau(z), S \setminus \{\tau(z)\}, V \setminus \{\tau(z)\}).$$

Lemma 15 (Deleted colors are preserved). *For every matched pair $z \rightsquigarrow \tau(z)$,*

$$z \in S \iff \tau(z) \in S, \quad z \in U \iff \tau(z) \in V.$$

Consequently

$$\tau(S) = S, \quad \tau(U) = V.$$

Proof sketch. The first color in the deleted card has size

$$|S| - \mathbf{1}_{z \in S}$$

on the left and

$$|S| - \mathbf{1}_{\tau(z) \in S}$$

on the right. Since η_z is color-preserving, these sizes are equal, so $z \in S$ if and only if $\tau(z) \in S$. The same argument for the second color uses $|U| = |V|$ and gives

$$|U| - \mathbf{1}_{z \in U} = |V| - \mathbf{1}_{\tau(z) \in V}.$$

The bijectivity of τ then gives $\tau(S) = S$ and $\tau(U) = V$. □

Extend η_z to a bijection

$$\hat{\eta}_z : V(K) \rightarrow V(K)$$

by sending z to $\tau(z)$. The previous lemma says

$$\hat{\eta}_z(S) = S, \quad \hat{\eta}_z(U) = V.$$

Moreover $\hat{\eta}_z$ preserves all graph adjacencies except possibly those incident to z . Define its fixed-host star error by

$$\text{Err}_z^K(\hat{\eta}_z) = \{x \in V(K) \setminus \{z\} : (zx \in E(K)) \not\leftrightarrow (\tau(z)\hat{\eta}_z(x) \in E(K))\}.$$

Lemma 16 (Zero fixed-host star error is orbit success). *If*

$$\text{Err}_z^K(\hat{\eta}_z) = \emptyset$$

for some fixed-host deck match, then U and V lie in the same $\text{Aut}(K, S)$ -orbit.

Proof sketch. The map η_z is already an isomorphism away from z . Empty star error says it also preserves every adjacency incident to z . Hence $\hat{\eta}_z \in \text{Aut}(K)$. It preserves S and carries U to V , so it is an element of $\text{Aut}(K, S)$ carrying U to V . □

Thus a fixed-host counterexample supplies a concrete object: a coherent family of color-correct one-star maps with no zero-error member. This is much sharper than the phrase “full-size orbit failure”: the failure is witnessed locally by star errors, one for each chosen deletion-card match.

Lemma 17 (Fixed-host rehosting). *Let $z \rightsquigarrow w$ be a fixed-host deck match with isomorphism*

$$\eta : (K - z, S \setminus \{z\}, U \setminus \{z\}) \cong (K - w, S \setminus \{w\}, V \setminus \{w\}).$$

Identify both deleted colored hosts with the common colored card

$$(L, S_L, U_L).$$

Then the two full colored hosts are recovered from L by adding one new vertex with the same pair of color memberships, and with attachment sets

$$N_K(z) \setminus \{z\}, \quad \eta^{-1}(N_K(w) \setminus \{w\}).$$

The attachment defect in this rehosted centered presentation is exactly

$$\text{Err}_z^K(\widehat{\eta}).$$

Proof sketch. The deleted-colors lemma gives

$$z \in S \iff w \in S, \quad z \in U \iff w \in V,$$

so the re-added vertex has the same first-color and second-color status on both sides. The deleted colored cards have already been identified by η , so the only remaining information needed to reconstruct the full colored host is the neighborhood of the re-added vertex in the deleted card. The difference between these two neighborhoods is precisely the star-error set. $\square$

This is the key exchange principle for the neighborhood-selection bottleneck. In the original centered problem, an attachment-color defect can be converted by a matched card into a one-star graph error. In the fixed-host problem, a second-color orbit defect can also be converted by a matched card into a one-star graph error, but now the deleted card carries both colors as common data. A full-size orbit failure would therefore have to be a closed system where these one-star errors never vanish and never descend.

Conjecture 13 (Fixed-host one-star descent). *Assume*

$$\text{Deck}_{K,S}(U) = \text{Deck}_{K,S}(V), \quad |U| = |V|,$$

and U, V are not in the same $\text{Aut}(K, S)$ -orbit. Then one can choose a fixed-host trade and a vertex $z \in U \triangle V$ such that the rehosted star error satisfies

$$|\text{Err}_z^K(\widehat{\eta}_z)| < |U \triangle V|.$$

For $|U \triangle V| = 2$, this conjecture asks for zero star error, hence an orbit-success automorphism. This is exactly the root-anchored base case needed after an exact edge-slide: the switch has moved from the original vertices p, q to a witness pair inside the direct recentered host, and the fixed-host descent must show that the new singleton switch cannot be a nonterminal sink.

Conjecture 14 (Fixed-host singleton switch). *Let*

$$U = T \cup \{a\}, \quad V = T \cup \{b\}, \quad a, b \notin T.$$

If

$$\text{Deck}_{K,S}(U) = \text{Deck}_{K,S}(V),$$

then some automorphism of K preserving S carries U to V .

This is the pure fixed-host analogue of single-switch rigidity. It rules out nonconjugate singleton moves inside the fixed background. Proving this singleton statement would settle the exact two-error endpoint of the root-anchored descent.

Remark 8 (Computational evidence for fixed-host singleton). *A brute-force check over all unlabeled hosts K on at most six vertices, all choices of S and T , and all ordered pairs $a, b \notin T$, found 433,448 fixed-host singleton instances with equal colored deletion decks. All 433,448 were orbit-successful, and in fact all 433,448 had a direct card edge*

$$(K - a, S \setminus \{a\}, T) \cong (K - b, S \setminus \{b\}, T).$$

No no-direct closed-flow candidate occurred in this range. This is stronger than needed for the inductive singleton package and is good evidence that detour selection, rather than neighborhood selection, is the core local obstruction.

A later random stress test on larger hosts sampled 60 random eight-vertex colored hosts and 25 random nine-vertex colored hosts, grouping all second-color subsets by their fixed-host deletion deck and extracting the singleton pairs. It found 1,024 equal-deck singleton pairs on eight vertices and 640 on nine vertices, again with no nondirect natural-shadow match. This is only evidence, but it supports the stronger direct-shadow target rather than merely the orbit conclusion.

5.11 Attacking the fixed-host singleton switch

Write

$$U = T \cup \{a\}, \quad V = T \cup \{b\}, \quad a, b \notin T.$$

The natural a -deleted fixed-host card is

$$H_a = (K - a, S \setminus \{a\}, T),$$

and the natural b -deleted card on the other side is

$$H_b = (K - b, S \setminus \{b\}, T).$$

If H_a appears on the V -side by deleting b , we have a direct fixed-host card edge $a \rightsquigarrow b$. If it appears by deleting a vertex $t \in T$, then the singleton switch has detoured through the common color:

$$(K - a, S \setminus \{a\}, T) \cong (K - t, S \setminus \{t\}, (T \setminus \{t\}) \cup \{b\}).$$

Lemma 18 (Fixed-host natural shadow isolation). *In the fixed-host singleton switch, suppose*

$$(K - a, S \setminus \{a\}, T)$$

is not isomorphic to any detour card

$$(K - t, S \setminus \{t\}, (T \setminus \{t\}) \cup \{b\}) \quad (t \in T).$$

Then there is a direct fixed-host card edge

$$(K - a, S \setminus \{a\}, T) \cong (K - b, S \setminus \{b\}, T).$$

Proof sketch. The card H_a occurs in the U -deck. By equality of fixed-host decks it must occur in the V -deck. The deleted-colors lemma says that, on the V -side, a card matching H_a must delete a vertex of $V = T \cup \{b\}$. By hypothesis it cannot delete a vertex of T , so it must delete b . $\square$

Thus the singleton switch again splits into detour selection and neighborhood selection. A direct edge gives an isomorphism

$$\eta : (K - a, S \setminus \{a\}, T) \cong (K - b, S \setminus \{b\}, T).$$

Extending η by $a \mapsto b$ gives a bijection preserving S and preserving $T \cup \{a\}$ as $T \cup \{b\}$. If its star error is empty, this is the desired automorphism. If its star error has size 2, it is again a single edge-slide at the moved vertex. So the exact two-error obstruction in the original graph has reproduced itself as a fixed-host singleton switch.

There is also a profile-pressure formulation. For a two-colored pattern P , let

$$m_P(x) = \pi_P(T \cup \{x\}) - \pi_P(T)$$

inside the fixed colored host (K, S) . This counts the contribution to P -occurrences caused by changing the second-color status of x , with the common background (K, S, T) subtracted away. It is best thought of as a finite-difference invariant: for some patterns it is a direct contribution, and for others it is a signed net contribution because occurrences containing x may change type when x becomes second-colored.

Lemma 19 (Moving-vertex profile balance). *Under fixed-host singleton deck equality,*

$$m_P(a) = m_P(b)$$

for every two-colored pattern P with fewer vertices than K .

Proof sketch. Proper pattern sums are deck-controlled, so

$$\pi_P(T \cup \{a\}) = \pi_P(T \cup \{b\})$$

for every proper pattern P . The term $\pi_P(T)$ depends only on the fixed background (K, S, T) , so subtracting it from both sides gives the displayed equality. $\square$

Write

$$\deg_R(x) = |N_K(x) \cap R|$$

for $R \subseteq V(K)$. The first consequences of moving-vertex profile balance are already strong.

Lemma 20 (Low-order singleton balances). *Under fixed-host singleton deck equality,*

$$a \in S \iff b \in S,$$

and

$$\deg_T(a) = \deg_T(b), \quad \deg_{K \setminus T}(a) = \deg_{K \setminus T}(b), \quad \deg_K(a) = \deg_K(b).$$

More generally, for each of the four background classes

$$S \cap T, \quad T \setminus S, \quad S \setminus T, \quad K \setminus (S \cup T),$$

the number of neighbors of a in that class equals the number of neighbors of b in that class.

Proof sketch. The first-color membership of the moving vertex is detected by the one-vertex pattern whose unique vertex is second-colored and first-colored: its finite difference is

$$\mathbf{1}_{x \in S}.$$

For the degree statements, use two-vertex edge patterns and read them as finite differences. The pattern in which both vertices are second-colored has finite difference

$$\deg_T(x).$$

The pattern in which exactly one vertex is second-colored has finite difference

$$\deg_{K \setminus T}(x) - \deg_T(x),$$

because edges from x to T were counted before x was second-colored and cease to have exactly one second-colored endpoint after x is added. Since the previous pattern already gives $\deg_T(a) = \deg_T(b)$, equality of the exactly-one-second-colored finite difference gives

$$\deg_{K \setminus T}(a) = \deg_{K \setminus T}(b).$$

Adding the two equalities gives $\deg_K(a) = \deg_K(b)$.

Refining the same two-vertex edge patterns by first-color status gives a triangular system whose entries are the neighbor counts in

$$S \cap T, \quad T \setminus S, \quad S \setminus T, \quad K \setminus (S \cup T).$$

The entries involving T are read from the both-second-colored patterns, and the entries outside T are then read from the exactly-one-second patterns after subtracting the known T -entries. All of these are proper pattern counts unless the host has at most two vertices, in which case the singleton statement is checked directly from the one-vertex cards. $\square$

Now suppose the natural a -shadow is absorbed by a detour vertex $t \in T$:

$$(K - a, S \setminus \{a\}, T) \cong (K - t, S \setminus \{t\}, (T \setminus \{t\}) \cup \{b\}).$$

Lemma 21 (Fixed-host absorber constraints). *Any such absorber $t \in T$ satisfies*

$$t \in S \iff a \in S \iff b \in S,$$

and

$$\deg_K(t) = \deg_K(a) = \deg_K(b).$$

Moreover, the second-color induced edge count gives

$$\deg_T(t) = \deg_T(b) - \mathbf{1}_{tb \in E(K)}.$$

If the common first-color status is 1, then the first-color induced edge count also gives

$$\deg_S(t) = \deg_S(a).$$

Proof sketch. The deleted-colors lemma applied to the match $a \rightsquigarrow t$ gives

$$a \in S \iff t \in S,$$

and comparing the size of the intersection of the two color classes in the deleted cards gives

$$t \in S \iff b \in S.$$

The uncolored edge counts of $K - a$ and $K - t$ give

$$\deg_K(a) = \deg_K(t),$$

and the low-order singleton balance gives $\deg_K(a) = \deg_K(b)$.

For the second-color edge equation, the left deleted card has second-color edge count $|E(K[T])|$. The right deleted card has second color $(T \setminus \{t\}) \cup \{b\}$, so its induced edge count is

$$|E(K[T])| - \deg_T(t) + \deg_{T \setminus \{t\}}(b).$$

Equating these and using

$$\deg_{T \setminus \{t\}}(b) = \deg_T(b) - \mathbf{1}_{tb \in E(K)}$$

gives the displayed formula. Finally, if $a, t \in S$, the first-color induced edge counts of $S \setminus \{a\}$ and $S \setminus \{t\}$ give $\deg_S(a) = \deg_S(t)$. $\square$

Combining the absorber equation with $\deg_T(a) = \deg_T(b)$, an absorber must satisfy

$$\deg_T(t) = \deg_T(a) - \mathbf{1}_{tb \in E(K)}.$$

So an absorber either has the same T -degree as a , if it is not adjacent to b , or exactly one less, if it is adjacent to b . This is a real restriction: detours can only pass through vertices of the same first-color status and nearly the same background degree profile as the moved vertex.

Corollary 1 (Degree-profile isolation). *If no vertex $t \in T$ satisfies the absorber constraints above, then the fixed-host singleton switch has a direct card edge $a \rightsquigarrow b$.*

Proof sketch. If the direct edge were absent, the natural shadow isolation lemma would force the natural a -deleted card to be absorbed by some $t \in T$. Such a vertex must satisfy the absorber constraints, contradicting the hypothesis. $\square$

There is a dual set of constraints for absorbers of the natural b -shadow on the other side. Thus a non-direct singleton counterexample would require two closed flows through T : one absorbing the a -deleted shadow and one absorbing the b -deleted shadow, both restricted to the same first-color class and to the same narrow degree-profile layer.

5.12 Fixed-host detour cycles

The trade-cycle language applies verbatim to the fixed-host singleton switch. Choose a fixed-host trade, so for each $z \in V(K)$ there is a matched deleted-card isomorphism

$$(K - z, S \setminus \{z\}, U \setminus \{z\}) \cong (K - \tau(z), S \setminus \{\tau(z)\}, V \setminus \{\tau(z)\}).$$

The deleted-colors lemma gives

$$\tau(S) = S, \quad \tau(U) = V.$$

In the singleton case $U = T \cup \{a\}$ and $V = T \cup \{b\}$, the edge-count cycle-balance therefore puts a and b on the same τ -cycle. More explicitly, along the directed arc

$$a = x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_k = b$$

from a to b , every internal vertex lies in T , provided the arc is chosen with no intermediate occurrence of a or b .

Lemma 22 (Fixed-host detour arcs are profile-flat). *Let*

$$a = x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_k = b$$

be the a -to- b arc of a fixed-host singleton trade, with internal vertices in T . Then every vertex on the arc has the same first-color status and the same total degree in K :

$$x_i \in S \iff a \in S, \quad \deg_K(x_i) = \deg_K(a) \quad (0 \leq i \leq k).$$

Proof sketch. For each trade edge $x_i \rightsquigarrow x_{i+1}$, the deleted first-color sizes give

$$x_i \in S \iff x_{i+1} \in S.$$

The uncolored edge counts of the matched deleted hosts give

$$|E(K)| - \deg_K(x_i) = |E(K)| - \deg_K(x_{i+1}),$$

so $\deg_K(x_i) = \deg_K(x_{i+1})$. Iterating along the arc gives the claim. $\square$

The absorber constraints sharpen this flatness at the first step: if $x_1 = t \in T$, then

$$\deg_T(t) = \deg_T(a) - \mathbf{1}_{tb \in E(K)}.$$

The dual constraints sharpen the final step into b . Thus a closed detour cycle is a string of vertices in T that are globally degree-flat and locally almost T -degree-flat, while also transporting the full deleted colored-card type. This is the object the no-closed-flow conjecture must rule out.

5.13 Absorber exchange maps

A detour absorber has more structure than just its deleted vertex. Suppose

$$\eta : (K - a, S \setminus \{a\}, T) \cong (K - t, S \setminus \{t\}, (T \setminus \{t\}) \cup \{b\})$$

with $t \in T$. Since η maps the second color T bijectively onto $(T \setminus \{t\}) \cup \{b\}$, there is a unique vertex

$$y_\eta \in T$$

such that

$$\eta(y_\eta) = b.$$

Call t the entry vertex of the absorber and y_η its exit vertex.

Extending η by $a \mapsto t$, the second-color transport has the form

$$a \mapsto t, \quad y_\eta \mapsto b,$$

with the remaining vertices of $T \setminus \{y_\eta\}$ mapped into $T \setminus \{t\}$. Thus a detour does not simply hide a in T . It injects a at the entry t , ejects the exit y_η to b , and permutes the rest of the common second color.

Definition 4 (Absorber exchange digraph). *For a fixed-host singleton state, define the a -absorber exchange digraph on T by drawing an edge*

$$t \longrightarrow y$$

whenever there is an absorber isomorphism with entry t and exit y . Define the b -absorber exchange digraph dually.

In this language, a no-direct closed flow is not just a set of absorbers; it is a circulation in these exchange digraphs. The entry vertex records where the outside color is inserted into T , and the exit vertex records where a vertex of T leaves to become the outside color on the other side.

Conjecture 15 (No closed absorber circulation). *In a fixed-host singleton state with equal colored deletion decks, the absorber exchange digraphs cannot support a closed circulation avoiding the direct edge $a \rightsquigarrow b$.*

This is a more combinatorial version of no closed fixed-host singleton flow. It suggests a possible proof strategy: assign a potential to vertices of T , built from their rooted profiles relative to (K, S, T, a, b) , and show that every absorber exchange edge strictly changes that potential in one direction unless a direct edge exists. The low-order absorber equations give the first coordinates of such a potential; the witness-charge equations give the next signed coordinates.

Lemma 23 (Exit constraints). *Let $t \rightarrow y$ be an absorber exchange edge, arising from*

$$\eta : (K - a, S \setminus \{a\}, T) \cong (K - t, S \setminus \{t\}, (T \setminus \{t\}) \cup \{b\})$$

with $\eta(y) = b$. Then

$$y \in S \iff b \in S,$$

$$\deg_K(y) - \mathbf{1}_{ay \in E(K)} = \deg_K(b) - \mathbf{1}_{tb \in E(K)},$$

and

$$\deg_T(y) = \deg_T(b) - \mathbf{1}_{tb \in E(K)}.$$

Proof sketch. The first statement follows because η preserves the first color and $\eta(y) = b$. The second statement compares the uncolored degrees of y in the deleted host $K - a$ and b in the deleted host $K - t$. The third compares the degrees of y and b inside the second-colored induced subgraphs of the two deleted colored cards. On the left this degree is $\deg_T(y)$; on the right it is

$$\deg_{T \setminus \{t\}}(b) = \deg_T(b) - \mathbf{1}_{tb \in E(K)}.$$

□

Corollary 2 (Closed exchanges avoid witness exits). *Suppose an absorber exchange edge $t \rightarrow y$ lies in a closed circulation, so that y is also usable as an absorber entry. Then*

$$\mathbf{1}_{ay \in E(K)} = \mathbf{1}_{by \in E(K)}.$$

In particular, $y \notin X^- \cup X^+$.

Proof sketch. Since y is also an absorber entry, the entry constraints give

$$\deg_K(y) = \deg_K(a) = \deg_K(b)$$

and

$$\deg_T(y) = \deg_T(a) - \mathbf{1}_{yb \in E(K)}.$$

The exit constraints and the low-order balance $\deg_T(a) = \deg_T(b)$ give

$$\mathbf{1}_{ay \in E(K)} = \mathbf{1}_{tb \in E(K)} \quad \text{and} \quad \mathbf{1}_{yb \in E(K)} = \mathbf{1}_{tb \in E(K)}.$$

Thus ay and by have the same adjacency status. $\square$

This is encouraging. The signed witnesses $X^- \cup X^+$ are exactly the vertices that the two-step charge needs in order to balance a non-twin singleton switch, but a closed absorber circulation cannot use such vertices as exits. Therefore a nonterminal witness cycle would have to carry its witness charges on vertices that are not the exchange exits, while the exchange exits themselves are a/b -neutral. That separation is a promising place to force a contradiction.

Conjecture 16 (Fixed-host exchange-chain shortening). *In a minimal no-direct fixed-host singleton state, the absorber exchange digraph has a closed circulation. Equivalently, if an absorber has an exit vertex $y \in T$ that is not usable as a further absorber entry, then one can modify the deck matching to obtain either a direct edge $a \rightsquigarrow b$ or a strictly shorter detour chain.*

This is the precise point where the proof could break. The exit constraints are strong only once exits close up into entries. If the deck can keep throwing exits into vertices that are not themselves absorbers, then the exchange-potential argument has no cycle to contradict. Thus the fixed-host detour campaign has two tasks: first prove exchange-chain shortening, then rule out closed absorber circulations using the exit and witness-charge constraints.

5.14 Natural-shadow fibers in the fixed-host singleton

There is also a multiplicity formulation, parallel to the earlier shadow-type flow. Let

$$H_a = (K - a, S \setminus \{a\}, T)$$

be the natural a -deleted card type. Define

$$L(H_a) = \{z \in V(K) : (K - z, S \setminus \{z\}, (T \cup \{a\}) \setminus \{z\}) \cong H_a\}$$

and

$$R(H_a) = \{z \in V(K) : (K - z, S \setminus \{z\}, (T \cup \{b\}) \setminus \{z\}) \cong H_a\}.$$

Deck equality gives

$$|L(H_a)| = |R(H_a)|.$$

The vertex a always lies in $L(H_a)$. The vertex b lies in $R(H_a)$ exactly when the direct natural shadow exists.

Since every card of type H_a has second-color size $|T|$, color-size comparison gives

$$L(H_a) \subseteq T \cup \{a\}, \quad R(H_a) \subseteq T \cup \{b\}.$$

Thus, if the direct edge is absent,

$$R(H_a) \subseteq T$$

and multiplicity balance becomes

$$|R(H_a) \cap T| = 1 + |L(H_a) \cap T|.$$

So a no-direct state has strictly more a -absorbers than a -suppliers. The dual natural shadow H_b gives the opposite-looking condition for the b -flow. A closed no-direct counterexample must reconcile these two imbalances inside the same finite set T .

The supplier constraints mirror the absorber constraints. A vertex $u \in T$ is an a -supplier for H_a if

$$(K - u, S \setminus \{u\}, (T \setminus \{u\}) \cup \{a\}) \cong (K - a, S \setminus \{a\}, T).$$

Then the same color-size and edge-count comparisons give

$$u \in S \iff a \in S, \quad \deg_K(u) = \deg_K(a),$$

and

$$\deg_T(u) = \deg_T(a) - \mathbf{1}_{ua \in E(K)}.$$

Compare this with the absorber condition

$$\deg_T(t) = \deg_T(a) - \mathbf{1}_{tb \in E(K)}.$$

Thus a vertex $z \in T$ that is neutral with respect to a and b , meaning

$$\mathbf{1}_{za \in E(K)} = \mathbf{1}_{zb \in E(K)},$$

passes the low-order T -degree test for being a supplier exactly when it passes the corresponding test for being an absorber. Only witnesses inside T can contribute to a low-order supplier–absorber imbalance.

Lemma 24 (Low-order imbalance is witness-supported). *Ignoring higher colored-card structure, the difference between the low-order candidate set for a -absorbers and the low-order candidate set for a -suppliers is supported on*

$$T \cap (X^- \cup X^+).$$

Proof sketch. The first-color and total-degree tests are the same for suppliers and absorbers. The only displayed low-order test that differs is the T -degree equation, and that equation differs only by replacing $\mathbf{1}_{za \in E(K)}$ with $\mathbf{1}_{zb \in E(K)}$. These indicators differ exactly on $X^- \cup X^+$. $\square$

This fits the exit constraint above: closed exchanges cannot use witnesses as exits, but the supplier–absorber imbalance can only be generated by witnesses. Therefore a closed no-direct flow would have to create its cardinality imbalance at witness entries while routing its exits through neutral vertices. That asymmetry is the current best candidate for a contradiction.

Conjecture 17 (Neutral absorber reversibility). *Let $t \in T$ be an a -absorber entry, and suppose t is neutral with respect to a, b :*

$$\mathbf{1}_{ta \in E(K)} = \mathbf{1}_{tb \in E(K)}.$$

Then either t is also an a -supplier for the natural shadow H_a , or there is a shorter detour chain, or there is a direct edge $a \rightsquigarrow b$.

This is a very small local reversibility statement. The absorber card replaces t by b in the second color. The supplier card replaces t by a . If t sees a and b in the same way, then the only reason absorption need not reverse to supply is a higher-order difference between the roles of a and b in the deleted card $K - t$. But such a difference should be visible in the proper moving-vertex profile unless it can be pushed further down the detour chain.

Conjecture 18 (No witness tail into a neutral circulation). *In a minimal no-direct fixed-host singleton state, an absorber exchange path cannot start at a witness entry*

$$t \in T \cap (X^- \cup X^+)$$

and then feed into a closed circulation all of whose vertices are a/b -neutral.

This is the refined obstruction. A closed neutral circulation by itself is not enough for a contradiction, because an excess absorber could appear as a tail leading into that circulation. The multiplicity imbalance has to be carried by witness entries, while the closed part of the exchange graph is neutral and reversible. Therefore the possible surviving shape is a witness-to-neutral tail.

Proposition 2 (Exchange reduction for direct shadow). *Assume fixed-host exchange-chain shortening, neutral absorber reversibility, and no witness tail into a neutral circulation. Then the fixed-host direct-shadow theorem holds.*

Proof sketch. Suppose the direct shadow fails in a minimal counterexample. By natural shadow isolation, the a -natural shadow is absorbed in T . Repeatedly follow absorber exchange exits. Exchange-chain shortening says that, unless a direct edge or a shorter detour appears, the path can be continued until it enters a closed circulation. Minimality excludes directness and shortening. On the closed part, the exit constraints force every vertex to be a/b -neutral; neutral reversibility makes the closed part contribute no net supplier-absorber imbalance.

The direct-shadow failure requires

$$|R(H_a) \cap T| = 1 + |L(H_a) \cap T|,$$

so some imbalance must be supplied outside the neutral reversible part. The low-order imbalance lemma says such imbalance is supported on witnesses. Thus the remaining obstruction is exactly a witness entry feeding into a neutral closed circulation, contradicting the no-witness-tail assumption. $\square$

This proposition isolates the current proof burden into three very local claims. Exchange-chain shortening is combinatorial: exits of absorbers must either close or shorten. Neutral reversibility is structural: an absorber at an a/b -neutral entry should be invertible as a supplier unless some smaller obstruction appears. The witness-tail lemma is the bridge between the two: it must show that a signed witness charge cannot disappear into a neutral reversible cycle.

Conjecture 19 (No supplier-absorber imbalance). *In a fixed-host singleton state with equal colored deletion decks, the two natural-shadow fiber balances cannot both avoid their direct vertices. More concretely, it is impossible to have*

$$b \notin R(H_a) \quad \text{and} \quad a \notin L(H_b).$$

This is another equivalent-looking form of the direct-shadow theorem. It may be more attackable because it isolates a signed cardinality imbalance: the a -flow needs one extra absorber in T , while the b -flow needs one extra absorber in the opposite direction. Any proof by exchange circulation should explain why these two imbalances cannot coexist.

5.15 The singleton type-flow graph

There is an even cleaner way to package the same multiplicity bookkeeping. For each $t \in T$, define the two card types

$$A_t = (K - t, S \setminus \{t\}, (T \setminus \{t\}) \cup \{a\}),$$

$$B_t = (K - t, S \setminus \{t\}, (T \setminus \{t\}) \cup \{b\}).$$

The second-color-size- $|T|$ part of the deck equality says

$$\{\{H_a\}\} + \{\{A_t : t \in T\}\} = \{\{H_b\}\} + \{\{B_t : t \in T\}\}.$$

Build a directed multigraph whose vertices are isomorphism classes of two-colored cards of this size, with one directed edge

$$B_t \longrightarrow A_t$$

for each $t \in T$, and one extra artificial edge

$$H_b \longrightarrow H_a.$$

Lemma 25 (Type-flow decomposition). *The singleton type-flow graph is balanced: every card type has equal indegree and outdegree. Consequently every edge lies in a directed cycle. In particular, if $H_a \neq H_b$, removing the artificial edge leaves a directed path*

$$H_a \rightsquigarrow H_b$$

using only edges $B_t \rightarrow A_t$ with $t \in T$.

Proof sketch. For a card type Y , its indegree in the graph is

$$|\{t : A_t \cong Y\}| + \mathbf{1}_{H_a \cong Y},$$

and its outdegree is

$$|\{t : B_t \cong Y\}| + \mathbf{1}_{H_b \cong Y}.$$

These are equal exactly by the displayed multiset equality. A finite balanced directed multigraph decomposes into directed cycles. The artificial edge $H_b \rightarrow H_a$ is in one such cycle; deleting it leaves a directed path from H_a to H_b . $\square$

This path is a canonical form of a no-direct detour. Its first edge is an absorber of H_a , because $H_a \cong B_t$ for the first t . Its final edge is a supplier into H_b , because $A_t \cong H_b$ for the last t . Intermediate edges are switches inside deleted T -cards:

$$(K - t, S \setminus \{t\}, (T \setminus \{t\}) \cup \{b\}) \rightsquigarrow (K - t, S \setminus \{t\}, (T \setminus \{t\}) \cup \{a\}).$$

Thus the no-direct problem is equivalent to ruling out nontrivial type-flow paths from the natural a -shadow to the natural b -shadow.

Conjecture 20 (No nontrivial singleton type-flow path). *In a fixed-host singleton state with equal colored deletion decks, the type-flow path from H_a to H_b must have length zero; equivalently,*

$$H_a \cong H_b.$$

This is just the fixed-host direct-shadow theorem in flow language, but it has two advantages. First, it is purely a finite directed multigraph statement attached to the equality of the second-color-size- $|T|$ deck slice. Second, it gives a canonical shortest-path parameter. A minimal counterexample can be chosen with the shortest possible nontrivial path

$$H_a = Y_0 \rightarrow Y_1 \rightarrow \cdots \rightarrow Y_r = H_b.$$

Any repeated card type on this path can be removed, so the intermediate types are distinct. The next proof attempt should show that a shortest path of positive length can be spliced or shortened using the absorber-exchange constraints above.

5.16 The complementary type-flow graph

The deck equality also has a second slice, obtained by deleting vertices that are not second-colored. Put

$$O = V(K) \setminus (T \cup \{a, b\}).$$

On the a -side, the vertices whose deletion leaves second-color size $|T| + 1$ are $O \cup \{b\}$; on the b -side they are $O \cup \{a\}$. Thus deck equality also gives

$$\{\{C_b^a\}\} + \{\{C_o^a : o \in O\}\} = \{\{C_a^b\}\} + \{\{C_o^b : o \in O\}\},$$

where

$$C_z^a = (K - z, S \setminus \{z\}, T \cup \{a\}), \quad C_z^b = (K - z, S \setminus \{z\}, T \cup \{b\}).$$

As before, this defines a balanced directed type-flow graph, now with edges

$$C_o^b \longrightarrow C_o^a \quad (o \in O)$$

and an artificial edge

$$C_a^b \longrightarrow C_b^a.$$

Lemma 26 (Complementary type-flow decomposition). *The complementary type-flow graph is balanced. Hence, if*

$$C_b^a \not\cong C_a^b,$$

then there is a directed path from C_b^a to C_a^b using only vertices deleted from O .

Proof sketch. The proof is the same indegree-outdegree count as for the singleton type-flow graph, applied to the second-color-size- $|T| + 1$ deck slice. $\square$

The endpoint cancellation

$$C_b^a \cong C_a^b$$

is also a direct recentering. It matches the card obtained from $(K, S, T \cup \{a\})$ by deleting the uncolored vertex b with the card obtained from $(K, S, T \cup \{b\})$ by deleting the uncolored vertex a . The re-added vertex has the same color data on both sides: it has the same first-color status by low-order singleton balance, and it is not second-colored on either side.

Lemma 27 (Complementary direct cancellation is a size drop). *Assume the colored centered extension theorem for centers with fewer than $|V(K)|$ vertices. If*

$$C_b^a \cong C_a^b,$$

then the fixed-host singleton switch is solved.

Proof sketch. Identify the two complementary endpoint cards with a common colored host L . Recenter the two colored full hosts by re-adding b on the a -side and a on the b -side. Since the matched card has one fewer vertex, the colored centered induction hypothesis applies to the resulting equal punctured decks over L . Transporting the resulting isomorphism back gives a color-preserving isomorphism

$$(K, S, T \cup \{a\}) \cong (K, S, T \cup \{b\}).$$

□

The low slice routes through T ; the high slice routes through O . A genuine no-direct singleton obstruction must make both slices balance. This is important because the witness set

$$X^- \cup X^+$$

is split between T and O . The low-slice supplier–absorber imbalance can only be generated by witnesses in T , while the complementary slice has the analogous imbalance supported on witnesses in O . Thus the two slices should not be studied separately: a counterexample would need compatible type-flow paths through both sides of the partition.

Conjecture 21 (Two-slice incompatibility). *There is no fixed-host singleton counterexample in which both the low type-flow path through T and the complementary type-flow path through O are nontrivial and shortest. At least one slice must admit a path shortening, a direct cancellation, or a size-drop recentering.*

Conjecture 22 (Two-slice direct cancellation). *In a fixed-host singleton state with equal colored deletion decks, at least one of the two endpoint cancellations holds:*

$$H_a \cong H_b \quad \text{or} \quad C_b^a \cong C_a^b.$$

This would be enough for the fixed-host singleton theorem by the two size-drop lemmas. The computational evidence actually supports the stronger first alternative $H_a \cong H_b$ in all cases checked, but the two-slice statement may be easier to prove because it allows either color-size slice to provide the direct recentering.

This is another way the proof might close. The low slice says excess absorber mass is supported by witnesses in T ; the high slice says the complementary excess is supported by witnesses in O . Since the total witness charge is globally balanced, these two flows should constrain each other. A future proof should make this precise by applying the two-step charge Q simultaneously to both color-size slices.

5.17 Two-slice charge balance

Define the signed witness charge

$$\delta(z) = \mathbf{1}_{bz \in E(K)} - \mathbf{1}_{az \in E(K)} \quad (z \notin \{a, b\}).$$

Thus $\delta = -1$ on X^- , $\delta = +1$ on X^+ , and $\delta = 0$ elsewhere. Degree balance gives the global cancellation

$$\sum_{z \notin \{a, b\}} \delta(z) = 0.$$

Equivalently,

$$\sum_{t \in T} \delta(t) = - \sum_{o \in O} \delta(o).$$

The same charge is visible directly on type-flow paths. For a low-slice card type Y with second color of size $|T|$, let

$$q(Y) = \sum_{x \in \text{second color of } Y} \deg_Y(x).$$

This is the second-color two-step invariant of that card.

Lemma 28 (Low path charge formula). *Let*

$$H_a = Y_0 \rightarrow Y_1 \rightarrow \cdots \rightarrow Y_r = H_b$$

be a low-slice type-flow path, where the edge $Y_{i-1} \rightarrow Y_i$ is indexed by $t_i \in T$. Then

$$q(Y_i) - q(Y_{i-1}) = \delta(t_i),$$

and therefore

$$\sum_{i=1}^r \delta(t_i) = q(H_b) - q(H_a) = \sum_{t \in T} \delta(t).$$

Proof sketch. The edge indexed by t compares

$$B_t = (K - t, S \setminus \{t\}, (T \setminus \{t\}) \cup \{b\})$$

with

$$A_t = (K - t, S \setminus \{t\}, (T \setminus \{t\}) \cup \{a\}).$$

The difference of the second-color two-step invariant between these two cards is exactly

$$\deg_{K-t}(a) - \deg_{K-t}(b) = \mathbf{1}_{bt \in E(K)} - \mathbf{1}_{at \in E(K)} = \delta(t),$$

using $\deg_K(a) = \deg_K(b)$. Summing along the path telescopes. Finally,

$$q(H_b) - q(H_a) = \sum_{t \in T} (\deg_{K-b}(t) - \deg_{K-a}(t)) = \sum_{t \in T} \delta(t).$$

□

There is an analogous formula for the complementary path through O , and its total charge is

$$\sum_{o \in O} \delta(o) = - \sum_{t \in T} \delta(t).$$

Thus the two type-flow paths carry opposite total two-step charge.

Conjecture 23 (No opposite shortest charge paths). *In a fixed-host singleton counterexample, one cannot have simultaneously a shortest positive low-slice path carrying charge*

$$\sum_{t \in T} \delta(t)$$

and a shortest positive complementary path carrying the opposite charge. At least one of the paths can be shortened or converted into a direct cancellation.

This conjecture is a charged version of two-slice incompatibility. If the charge on one slice is nonzero, then any path in that slice must pass through signed witnesses. The other slice must carry exactly the opposite signed charge. If both paths were shortest and nontrivial, the two paths would give two incompatible ways of transporting the same global witness imbalance. If the slice charge is zero, then q alone cannot see the obstruction and one must repeat the same argument with higher finite-difference pattern invariants.

5.18 The rectangular second boundary

The two-slice picture has a more exact form. The low slice deletes either the moving vertex or a vertex of T . The complementary slice deletes either the other moving vertex or a vertex of

$$O = V(K) \setminus (T \cup \{a, b\}).$$

If we then delete one more vertex from the opposite side of the partition, both slices see the same two-hole cards.

For $t \in T$ and $o \in O$, define

$$D_{t,o}^a = (K - \{t, o\}, S \setminus \{t, o\}, (T \setminus \{t\}) \cup \{a\}),$$

$$D_{t,o}^b = (K - \{t, o\}, S \setminus \{t, o\}, (T \setminus \{t\}) \cup \{b\}).$$

Then the elementary square identities are

$$A_t - o = C_o^a - t = D_{t,o}^a, \quad B_t - o = C_o^b - t = D_{t,o}^b.$$

Let ∂_R denote the fixed-host partial deletion operator that deletes only vertices in R from the displayed representatives, and then passes to colored isomorphism classes. This operator is not an invariant of an abstract card type unless R has been named in the ambient host. That is fine here: the following identity is a fixed-host identity before taking isomorphism classes.

Lemma 29 (Rectangular second-boundary identity). *In a fixed-host singleton state with equal colored deletion decks,*

$$\partial_O([H_b] - [H_a]) = \partial_T([C_a^b] - [C_b^a])$$

and both sides are equal to the rectangular chain

$$\sum_{t \in T} \sum_{o \in O} ([D_{t,o}^a] - [D_{t,o}^b]).$$

Proof sketch. The low slice gives

$$[H_b] - [H_a] = \sum_{t \in T} ([A_t] - [B_t]).$$

Deleting the vertices of O from this identity gives

$$\partial_O([H_b] - [H_a]) = \sum_{t \in T} \sum_{o \in O} ([A_t - o] - [B_t - o]) = \sum_{t \in T} \sum_{o \in O} ([D_{t,o}^a] - [D_{t,o}^b]).$$

The complementary slice gives

$$[C_a^b] - [C_b^a] = \sum_{o \in O} ([C_o^a] - [C_o^b]).$$

Deleting the vertices of T from this identity gives the same double sum, using $C_o^a - t = D_{t,o}^a$ and $C_o^b - t = D_{t,o}^b$. $\square$

This is stronger than the scalar charge balance. The two-step charge q is obtained by applying one numerical functional to the rectangular chain. Higher finite-difference pattern invariants are obtained by applying other colored subgraph-count functionals to the same chain. Thus the right object is not a single charge but the whole rectangular boundary on $T \times O$.

Conjecture 24 (No nonzero rectangular kernel). *Let $X_a = (K, S, T \cup \{a\})$ and $X_b = (K, S, T \cup \{b\})$ be a fixed-host singleton pair with equal colored deletion decks. If both endpoint differences*

$$[H_b] - [H_a], \quad [C_a^b] - [C_b^a]$$

are nonzero, then their rectangular second boundary cannot agree as in the previous lemma. Equivalently, the rectangular identity forces at least one of

$$H_a \cong H_b, \quad C_b^a \cong C_a^b.$$

This is the cleanest current form of the two-slice direct-cancellation campaign. A counterexample would be a nonzero rectangular kernel element: two different endpoint defects whose named second deletions across the opposite side of the partition are indistinguishable. That is a much more specific obstruction than an arbitrary homology class of the deletion complex.

The important nuance is that the rectangular chain is the common image of two endpoint defects, not the defect itself. A moment of the $D_{t,o}^*$ cards only studies this common image. To kill the kernel, one has to show that a nonzero endpoint defect cannot be invisible to its named partial boundary.

Conjecture 25 (Full-support rectangular squeeze). *Work in a minimal fixed-host singleton counterexample, minimizing first $|V(K)|$ and then the total length of the low and complementary type-flow paths. Assume both endpoint cancellations fail. Let P be a smallest colored pattern that distinguishes the two low endpoints H_a and H_b . If a copy of P responsible for the difference misses some vertex of O , then the matching term in $\partial_T([C_a^b] - [C_b^a])$ produces either a smaller fixed-host singleton obstruction, a path shortening, or a direct cancellation. Hence, in a minimal unresolved counterexample, every such low separator must use all vertices of O . Dually, every complementary separator must use all vertices of T . These two full-support requirements are incompatible with the original one-vertex deck equality.*

This is now the best candidate for the “make or break” lemma. If it is true, the fixed-host singleton theorem follows by the two size-drop lemmas. If it is false, the strategy has not merely failed to find the right charge: there is a genuine finite rectangular kernel, and the centered-extension campaign needs a new ingredient that is not just another Kelly-style moment.

The intuition is simple. The partial boundary $\partial_O([H_b] - [H_a])$ detects every low-endpoint discrepancy that survives after deleting at least one vertex of O . Such a visible discrepancy is not automatically a contradiction, because the rectangular identity says that it is matched by the complementary partial boundary. But that match is a two-hole card isomorphism, and therefore should be exactly the kind of smaller detour from which we can extract a path shortening or a size drop. If no such extraction is possible, the first visible difference must be supported on all of O . The dual argument forces the complementary first difference to be supported on all of T . A genuine kernel would have to place the first visible difference simultaneously on both full supports. That is a very small corner of the search space, and it is the corner to attack next.

Remark 9 (Candidate proof of the squeeze). *Assume a minimal counterexample to the fixed-host singleton theorem in which both endpoint cancellations fail.*

1. *By the smaller colored centered theorem, the nonisomorphic low endpoints H_a, H_b cannot have identical colored decks. Choose a deleted low-endpoint card type Z whose multiplicity differs, minimizing the size of a colored pattern detecting the difference.*

2. If Z arises by deleting a vertex of O , then Z appears in $\partial_O([H_b] - [H_a])$. The rectangular identity matches this occurrence inside $\partial_T([C_a^b] - [C_b^a])$. Unwinding the matched two-hole card gives a smaller fixed-host singleton state over $K - \{t, o\}$, or else an isomorphism that splices one of the two type-flow paths. Minimality rules this out unless it is already a direct cancellation.
3. Therefore every first low deck discrepancy must come from deleting a vertex of T . Symmetrically, every first complementary deck discrepancy must come from deleting a vertex of O .
4. Repeat the argument one level down. A low discrepancy obtained by deleting $t \in T$ is a singleton difference inside the smaller host $K - t$. If any detecting copy misses a vertex of O , the same rectangular matching produces a smaller obstruction or a path shortening. Hence the detecting copy must use all of O .
5. The dual argument forces the complementary detecting copy to use all of T . These two full-support detectors together determine a proper colored pattern in the original full hosts X_a, X_b , contradicting their equal deck-reconstructed proper pattern counts.

The fragile step is the second one: a matched two-hole card must really produce a smaller singleton obstruction or a valid splice. This is the exact place where the approach can break. The good news is that it is now a finite, local statement about two deleted vertices, not a global statement about arbitrary graph reconstruction.

5.19 The two-hole splice lemma

We now isolate the local statement hidden in the previous paragraph. A rectangular cancellation pairs a positive two-hole card with a negative two-hole card. Thus its basic datum is

$$t, t' \in T, \quad o, o' \in O, \quad \theta : D_{t,o}^a \cong D_{t',o'}^b.$$

Unpacked, θ is a color-preserving isomorphism

$$(K - \{t, o\}, S \setminus \{t, o\}, (T \setminus \{t\}) \cup \{a\}) \cong (K - \{t', o'\}, S \setminus \{t', o'\}, (T \setminus \{t'\}) \cup \{b\}).$$

Definition 5 (Two-hole ports). *For such a two-hole match θ , define its low ports by looking at the second color:*

$$\theta(a) \in (T \setminus \{t'\}) \cup \{b\}, \quad \theta^{-1}(b) \in (T \setminus \{t\}) \cup \{a\}.$$

If $\theta(a) \neq b$, call $\theta(a)$ the low exit. If $\theta^{-1}(b) \neq a$, call $\theta^{-1}(b)$ the low entry.

Define its complementary ports by looking at the vertices that are not second-colored:

$$\theta(b) \in (O \setminus \{o'\}) \cup \{a\}, \quad \theta^{-1}(a) \in (O \setminus \{o\}) \cup \{b\}.$$

If $\theta(b) \neq a$, call $\theta(b)$ the complementary exit. If $\theta^{-1}(a) \neq b$, call $\theta^{-1}(a)$ the complementary entry.

Lemma 30 (Two-hole color bookkeeping). *Every two-hole match has exactly the four alternatives described above: the image of a is either the moved second-colored vertex b or a vertex of $T \setminus \{t'\}$; the preimage of b is either a or a vertex of $T \setminus \{t\}$; the image of b is either a or a vertex of $O \setminus \{o'\}$; and the preimage of a is either b or a vertex of $O \setminus \{o\}$.*

Proof sketch. The first two statements are exactly preservation of the second color. The last two use preservation of the complement of the second color, together with the fact that $a, b \notin T$ and $o, o' \in O$. $\square$

The productive cases are the ones where a two-hole match identifies an endpoint with its counterpart, or gives a chord in one of the type-flow paths. For example, if $\theta(a) = b$, then the match respects the moved second-colored vertex. If also $\theta(b) = a$, then it respects the moved uncolored vertex as well; the only remaining ambiguity is how to reinsert the two deleted holes t, o and t', o' . This is a two-star error, hence a smaller centered problem over the common two-hole card. If one of the low ports lies in T , then the match says that the low type-flow path can enter or leave the same card type through an internal T -vertex; in a shortest path, this should create a chord. The complementary ports do the same for the high path through O .

Conjecture 26 (Two-hole splice lemma). *Work in a minimal fixed-host singleton counterexample, minimizing $|V(K)|$ and then the total length of the chosen low and complementary type-flow paths. Let*

$$\theta : D_{t,o}^a \cong D_{t',o'}^b$$

be a two-hole match appearing in the rectangular boundary matching. Then at least one of the following holds.

1. *The match is endpoint-compatible:*

$$\theta(a) = b, \quad \theta(b) = a.$$

Re-inserting the two deleted holes either gives a direct endpoint cancellation or a smaller colored centered state over the two-hole card.

2. *The low ports define a chord in the low type-flow path, so the path from H_a to H_b can be shortened.*
3. *The complementary ports define a chord in the complementary type-flow path, so the path from C_b^a to C_a^b can be shortened.*
4. *The match exposes a smaller fixed-host singleton obstruction inside one of the deleted-hole cards, contradicting minimality of $|V(K)|$.*

Equivalently, a minimal unresolved counterexample would have to make the remaining rectangular matches *essentially locked*: neither moved vertex maps to its counterpart, all four ports are internal, and the induced low and complementary port relations give no chord in either shortest path.

Conjecture 27 (No essential locked two-hole match). *There is no essential locked two-hole match in a minimal fixed-host singleton counterexample, after all direct cancellations, smaller-obstruction moves, and path-splicing chords have been removed from the rectangular matching.*

This is the most concrete local obstruction currently visible. A locked match would have data

$$\begin{aligned} \theta(a) &= x_T \in T \setminus \{t'\}, & \theta^{-1}(b) &= y_T \in T \setminus \{t\}, \\ \theta(b) &= x_O \in O \setminus \{o'\}, & \theta^{-1}(a) &= y_O \in O \setminus \{o\}, \end{aligned}$$

with no shortening relation among the corresponding low and complementary flow vertices. The color bookkeeping already forces

$$a \in S \iff x_T \in S \iff y_T \in S \iff b \in S$$

and

$$b \in S \iff x_O \in S \iff y_O \in S \iff a \in S.$$

Degree and two-step balance should add the same equal-degree and witness neutrality constraints as in the absorber analysis. Thus a locked match is not an arbitrary two-hole isomorphism; it is a highly constrained four-port near-automorphism of the host with two stars removed.

Remark 10 (Locked maps are not forbidden absolutely). *The adjective “essential” matters. Locked two-hole isomorphisms can occur in already solved states. For example, in the empty six-vertex graph with $S = \emptyset$, $T = \{0, 1\}$, $a = 2$, $b = 3$, and $t = t' = 0$, $o = o' = 4$, the two-hole cards are just colored empty graphs. A color-preserving bijection can send a to the remaining T -vertex and b to an O -vertex, making all ports internal. But the singleton switch is trivially solved by a direct automorphism. Therefore the target is not to prohibit locked local maps; it is to show that no locked map can be the last surviving term in a minimal unresolved rectangular kernel.*

The same warning applies even after extension errors are included. A random seven-vertex solved singleton instance produced a locked two-hole map with nonzero low and high extension errors, but with the low ports equal and the complementary ports equal. Such a map is not essential: the port loops are exactly the local splice signal. This reinforces the intended target. The forbidden object is a locked map with nonzero unresolved errors and no port-loop, no path chord, no direct cancellation, and no smaller obstruction.

Lemma 31 (First-order equations for a locked match). *Let $\theta : D_{t,o}^a \cong D_{t',o'}^b$ be locked, with ports*

$$\theta(a) = x_T, \quad \theta(y_T) = b, \quad \theta(b) = x_O, \quad \theta(y_O) = a.$$

Then the following total-degree equations hold:

$$\begin{aligned} \deg_K(a) - \mathbf{1}_{at} - \mathbf{1}_{ao} &= \deg_K(x_T) - \mathbf{1}_{x_T t'} - \mathbf{1}_{x_T o'}, \\ \deg_K(y_T) - \mathbf{1}_{y_T t} - \mathbf{1}_{y_T o} &= \deg_K(b) - \mathbf{1}_{b t'} - \mathbf{1}_{b o'}, \\ \deg_K(b) - \mathbf{1}_{bt} - \mathbf{1}_{bo} &= \deg_K(x_O) - \mathbf{1}_{x_O t'} - \mathbf{1}_{x_O o'}, \\ \deg_K(y_O) - \mathbf{1}_{y_O t} - \mathbf{1}_{y_O o} &= \deg_K(a) - \mathbf{1}_{at'} - \mathbf{1}_{ao'}. \end{aligned}$$

The second-color degree equations are

$$\begin{aligned} \deg_T(a) - \mathbf{1}_{at} &= \deg_T(x_T) - \mathbf{1}_{x_T t'} + \mathbf{1}_{x_T b}, \\ \deg_T(y_T) - \mathbf{1}_{y_T t} + \mathbf{1}_{y_T a} &= \deg_T(b) - \mathbf{1}_{b t'}, \\ \deg_T(b) - \mathbf{1}_{bt} + \mathbf{1}_{ba} &= \deg_T(x_O) - \mathbf{1}_{x_O t'} + \mathbf{1}_{x_O b}, \end{aligned}$$

and

$$\deg_T(y_O) - \mathbf{1}_{y_O t} + \mathbf{1}_{y_O a} = \deg_T(a) - \mathbf{1}_{at'} + \mathbf{1}_{ab}.$$

Proof sketch. The total-degree equations are obtained by comparing degrees in the deleted uncolored hosts $K - \{t, o\}$ and $K - \{t', o'\}$. The second-color equations compare the number of neighbors of the given vertex inside the second-color class. On the domain side the second color is $(T \setminus \{t\}) \cup \{a\}$; on the codomain side it is $(T \setminus \{t'\}) \cup \{b\}$. Substituting the four port images gives the four displayed formulas. $\square$

These equations are deliberately elementary. They are the first Lean-facing normal form for the locked case: all terms are degrees in K plus adjacency indicators among the ten named vertices

$$a, b, t, t', o, o', x_T, y_T, x_O, y_O,$$

with possible coincidences among the ports. The low-order singleton balance can then replace $\deg_K(a)$ by $\deg_K(b)$ and $\deg_T(a)$ by $\deg_T(b)$. What remains is a finite system of $\{0, 1\}$ -corrections. A locked match is essential only if this system is compatible while still producing no port equality, no chord, and no smaller singleton state.

Definition 6 (Two-hole extension errors). *Let $\theta : D_{t,o}^a \cong D_{t',o'}^b$. The low extension error is*

$$\text{Err}_O(\theta) = \theta(N_K(o) \setminus \{t, o\}) \triangle (N_K(o') \setminus \{t', o'\}),$$

viewed as a subset of $V(K) \setminus \{t', o'\}$. The high extension error is

$$\text{Err}_T(\theta) = \theta(N_K(t) \setminus \{t, o\}) \triangle (N_K(t') \setminus \{t', o'\}).$$

The notation reflects which deleted hole is being reinserted. The low error measures whether θ extends after re-inserting the two O -holes o, o' , hence whether it gives an isomorphism between the low one-hole cards A_t and $B_{t'}$. The high error measures whether θ extends after re-inserting the two T -holes t, t' , hence whether it gives an isomorphism between the complementary one-hole cards C_o^a and $C_{o'}^b$.

It is useful to keep signs, not just supports. For $z \in V(K) \setminus \{t, o\}$, define

$$\sigma_O^\theta(z) = \mathbf{1}_{oz \in E(K)} - \mathbf{1}_{o'\theta(z) \in E(K)},$$

and

$$\sigma_T^\theta(z) = \mathbf{1}_{tz \in E(K)} - \mathbf{1}_{t'\theta(z) \in E(K)}.$$

Then

$$|\text{Err}_O(\theta)| = \sum_z |\sigma_O^\theta(z)|, \quad |\text{Err}_T(\theta)| = \sum_z |\sigma_T^\theta(z)|.$$

The sign records whether the domain reinserted hole has an extra adjacency (+1) or the codomain reinserted hole has an extra adjacency (−1).

Lemma 32 (Zero extension error gives a chord). *Let $\theta : D_{t,o}^a \cong D_{t',o'}^b$.*

1. *If $o \in S \iff o' \in S$ and $\text{Err}_O(\theta) = \emptyset$, then θ extends by $o \mapsto o'$ to a colored isomorphism*

$$A_t \cong B_{t'}.$$

Thus, if A_t and $B_{t'}$ occur as nonconsecutive vertices of the chosen low type-flow path, that path has a chord and can be shortened.

2. *If $t \in S \iff t' \in S$ and $\text{Err}_T(\theta) = \emptyset$, then θ extends by $t \mapsto t'$ to a colored isomorphism*

$$C_o^a \cong C_{o'}^b.$$

Thus, if C_o^a and $C_{o'}^b$ occur as nonconsecutive vertices of the chosen complementary type-flow path, that path has a chord and can be shortened.

Proof sketch. For the first statement, the only graph adjacencies not already checked by θ are the adjacencies incident to the reinserted O -hole. These are exactly compared by $\text{Err}_O(\theta)$. The condition $o \in S \iff o' \in S$ says that the reinserted vertices have the same first-color status; both are outside the second color. The second statement is identical, with t, t' reinserted instead. In that case both reinserted vertices are second-colored, so only first-color compatibility is additional. $\square$

Therefore an essential locked match must satisfy four negative conditions: the two endpoint maps $a \mapsto b$ and $b \mapsto a$ fail, the low extension either has nonzero error or wrong first-color status at o, o' , and the high extension either has nonzero error or wrong first-color status at t, t' . The locked case is not just a four-port condition; it is a simultaneous pair of nonzero one-star extension errors over the same two-hole card.

Lemma 33 (Extension-error size tests). *For a two-hole match $\theta : D_{t,o}^a \cong D_{t',o'}^b$, put*

$$A_O = N_K(o) \setminus \{t, o\}, \quad B_O = N_K(o') \setminus \{t', o'\},$$

and

$$A_T = N_K(t) \setminus \{t, o\}, \quad B_T = N_K(t') \setminus \{t', o'\}.$$

Then

$$|\theta(A_O)| - |B_O| = \deg_K(o) - \mathbf{1}_{o \in E(K)} - \deg_K(o') + \mathbf{1}_{o' \in E(K)},$$

and

$$|\theta(A_T)| - |B_T| = \deg_K(t) - \mathbf{1}_{t \in E(K)} - \deg_K(t') + \mathbf{1}_{t' \in E(K)}.$$

In particular, if the corresponding reinserted one-hole cards have the same edge count, then the relevant extension error has even cardinality.

Proof sketch. The first two displayed identities just count the attachment sets of the reinserted vertices into the common two-hole card. The difference between the edge counts of the low one-hole cards A_t and $B_{t'}$ is exactly the first displayed quantity; the difference between the edge counts of the complementary one-hole cards C_o^a and $C_{o'}^b$ is exactly the second. If the two attachment sets have the same cardinality, then

$$|A \Delta B| = 2|A \setminus B|$$

after transporting one set by θ , so the extension error is even. $\square$

Thus a one-edge locked error is immediately visible as an edge-count imbalance between the corresponding one-hole cards. In a shortest type-flow obstruction, such an imbalance must be compensated elsewhere along the same path. This suggests a second potential: sum the sizes of the low and high extension errors over a rectangular matching. Any productive splice reduces this potential to zero on one rectangle; an essential locked matching would have to keep the potential positive while maintaining all edge-count telescoping identities.

Definition 7 (Balanced residual rectangular potential). *The rectangular chain*

$$R_{T,O} = \sum_{t \in T} \sum_{o \in O} ([D_{t,o}^a] - [D_{t,o}^b])$$

need not be the zero chain. Its nonzero coefficients are exposed rectangular boundary terms. After removing all exposed terms that can be used for a full-support squeeze or a smaller obstruction, a remaining balanced residual subchain has, for each isomorphism class of two-hole cards, equally many positive and negative occurrences.

On such a balanced residual subchain, choose a bijective matching between the positive occurrences $D_{t,o}^a$ and the negative occurrences $D_{t',o'}^b$, together with a specific isomorphism θ for each matched pair. Define

$$\Phi = \sum_{\theta} (|\text{Err}_O(\theta)| + |\text{Err}_T(\theta)|).$$

Among all such residual rectangular matchings, choose one minimizing Φ . Subject to that, minimize the number of essential locked matches.

Remark 11 (Rigidified vector form). Fix an isomorphism class $\mathcal{D}$ of two-hole cards, and choose a representative colored card R . For every positive occurrence

$$D_{t,o}^a \cong R,$$

choose one identification $\iota : D_{t,o}^a \cong R$. This occurrence then determines two attachment vectors on $V(R)$:

$$u_O(z) = \mathbf{1}_{o\iota^{-1}(z) \in E(K)}, \quad u_T(z) = \mathbf{1}_{t\iota^{-1}(z) \in E(K)}.$$

Similarly every negative occurrence $D_{t',o'}^b \cong R$ determines two attachment vectors

$$v_O(z) = \mathbf{1}_{o'\iota'^{-1}(z) \in E(K)}, \quad v_T(z) = \mathbf{1}_{t'\iota'^{-1}(z) \in E(K)}.$$

After this rigidification, choosing an isomorphism $\theta : D_{t,o}^a \cong D_{t',o'}^b$ is the same as choosing compatible identifications with R , and its contribution to Φ is the Hamming transport cost

$$d(u_O, v_O) + d(u_T, v_T).$$

Zero cost in the O -coordinate means the low one-hole cards match. Zero cost in the T -coordinate means the complementary one-hole cards match.

This vector form clarifies the nature of the remaining problem. The rectangular boundary does not merely say that two multisets of card types are equal. For each two-hole card type R , it gives two finite multisets of pairs of binary attachment vectors

$$(u_O, u_T) \quad \text{and} \quad (v_O, v_T).$$

The direct-cancellation campaign first has to deal with exposed unbalanced two-hole terms. Only after those terms are removed or converted into smaller obstructions does one get a balanced residual transport problem. In that residual problem, an essential locked rectangular kernel is an optimal transport plan in which every matched pair differs in both coordinates and none of those differences gives a path chord.

Remark 12 (The transport statement is not abstractly true). The last sentence is not a theorem about arbitrary multisets of binary-vector pairs. Abstractly, one can have a positive multiset containing a single pair (u_O, u_T) and a negative multiset containing a single pair (v_O, v_T) with $u_O \neq v_O$ and $u_T \neq v_T$; then every matching has positive cost in both coordinates. Therefore the missing argument must use the extra structure: these vectors arise as the two deleted-hole attachment vectors inside one fixed graph, they sit in the low and complementary type-flow paths, and their aggregate comes from equality of an actual one-vertex deletion deck. This is a useful guardrail for the Lean formalization: the minimum-error lemma cannot be stated as a standalone transport theorem, and it only applies after the exposed rectangular boundary terms have been handled.

Definition 8 (Active rectangle of two shortest paths). *Fix a shortest positive low type-flow path*

$$H_a = Y_0 \rightarrow Y_1 \rightarrow \cdots \rightarrow Y_r = H_b,$$

where the edge $Y_{i-1} \rightarrow Y_i$ is indexed by $t_i \in T$. Also fix a shortest positive complementary type-flow path

$$C_b^a = Z_0 \rightarrow Z_1 \rightarrow \cdots \rightarrow Z_s = C_a^b,$$

where the edge $Z_{j-1} \rightarrow Z_j$ is indexed by $o_j \in O$. The active rectangle is the multiset of pairs

$$\{1, \dots, r\} \times \{1, \dots, s\},$$

whose (i, j) -entry is the two-hole pair

$$D_{t_i, o_j}^a, \quad D_{t_i, o_j}^b.$$

Lemma 34 (Active rectangular identity). *For the two chosen paths,*

$$\sum_{i=1}^r \sum_{j=1}^s ([D_{t_i, o_j}^a] - [D_{t_i, o_j}^b])$$

is obtained both by deleting the complementary path holes o_j from the low path telescoping identity and by deleting the low path holes t_i from the complementary path telescoping identity.

Proof sketch. The low path gives the chain identity

$$[H_b] - [H_a] = \sum_{i=1}^r ([A_{t_i}] - [B_{t_i}]).$$

Deleting each o_j and summing over j gives the displayed double sum. The complementary path gives

$$[C_b^a] - [C_a^b] = \sum_{j=1}^s ([C_{o_j}^a] - [C_{o_j}^b]).$$

Deleting each t_i and summing over i gives the same double sum, since

$$A_{t_i} - o_j = C_{o_j}^a - t_i = D_{t_i, o_j}^a, \quad B_{t_i} - o_j = C_{o_j}^b - t_i = D_{t_i, o_j}^b.$$

□

This active rectangle is the right object for the splice lemma, but it has the same caveat as the full rectangle: it need not be balanced as a positive-negative chain. If it has an exposed unbalanced two-hole type, that type is already a candidate separator. If an active balanced residual remains, then a zero O -extension error in a matched entry gives an isomorphism

$$A_{t_i} \cong B_{t_{i'}}$$

for some active i, i' , hence a chord or endpoint cancellation in the low path. A zero T -extension error similarly gives a chord or endpoint cancellation in the complementary path. Rectangular matches outside the active rectangle may be harmless side cycles; they should not be part of the minimum-error obstruction.

Conjecture 28 (Minimum-error residual matching has no essential lock). *In a minimal fixed-host singleton counterexample, after all exposed rectangular boundary terms have been converted into full-support squeezes or smaller obstructions, a minimum-error matching of the balanced residual rectangular chain contains no essential locked two-hole match.*

This is a more operational version of the two-hole splice lemma. If a locked match has nonzero low or high extension error, then some adjacency incident to one of the deleted holes is mismatched. Since the total rectangular chain is balanced, the same mismatched adjacency must be corrected by another occurrence of the same two-hole card type. Swapping the partners along the resulting alternating correction cycle should either reduce Φ , create a zero-error extension in one direction, or expose a smaller singleton obstruction. In a Φ -minimal matching, none of these can happen, so an essential locked match should be impossible.

Conjecture 29 (Locked correction cycle). *Let a minimum-error rectangular matching contain an essential locked match θ . Choose a signed low or high error, say $\sigma_O^\theta(z) = +1$. Then the rectangular balance supplies a finite alternating sequence of matched two-hole cards carrying the same named adjacency discrepancy with signs alternating $+1, -1, +1, \dots, -1$. Swapping partners around this correction cycle strictly decreases*

$$\sum_{\theta} \sum_z (|\sigma_O^\theta(z)| + |\sigma_T^\theta(z)|),$$

unless one of the swaps creates a zero-extension chord or a smaller singleton obstruction.

This conjecture is the combinatorial heart of the minimum-error approach. It is the rectangular analogue of cancelling opposite signs in an optimal matching problem. The nontrivial point is that the signs are attached to named adjacencies involving the deleted holes, while the two-hole cards are unlabelled up to isomorphism. The two-hole isomorphisms must transport the names coherently enough for the alternating cycle to be defined.

Remark 13 (Why this is Lean-shaped). *The potential Φ is finite and purely combinatorial. A Lean proof would not need to reason about arbitrary choices forever: choose a minimizing rectangular matching by finite well-ordering, then prove that any locked match admits a local partner-swap decreasing the lexicographic potential*

$$(\Phi, \# \text{essential locks}).$$

This is exactly the same architecture as the earlier minimal-defect strategy, but now the moving part is a finite matching of two-hole cards rather than an unrooted deck matching.

Remark 14 (Next local attack). *The immediate task is now to prove that locked matches cannot exist. The first pass should use only first-order data: first-color status, total degree, degree into T , degree into O , and the signed witness charge δ . If these invariants force*

$$x_T = y_T \quad \text{or} \quad x_O = y_O,$$

then the corresponding port relation is a loop or a chord, hence a splice. If they do not, the locked case becomes a finite four-port configuration whose second-order boundary can be searched directly.

Remark 15 (Lean readiness of the fixed-host block). *The following parts of the fixed-host singleton campaign are now formalization-ready definitions or straightforward lemmas.*

1. The fixed-host colored deletion deck $\text{Deck}_{K,S}(U)$.
2. The low and complementary type-flow decompositions.
3. The rectangular second-boundary identity.
4. Two-hole ports and the color-bookkeeping lemma.
5. The first-order locked-match equations.
6. The extension-error sets $\text{Err}_O, \text{Err}_T$, their signed versions σ_O, σ_T , and the zero-extension-error chord lemma.
7. The rigidified vector form for each two-hole card type, and the finite potential Φ for a rectangular matching.

The part that is not yet Lean-ready as a proof is the correction-cycle descent: from an essential locked match in a Φ -minimal rectangular matching, construct a partner swap that reduces $(\Phi, \#\text{essential locks})$, or else extract a smaller singleton obstruction. Once that local descent is proved in English, the fixed-host singleton theorem has a plausible Lean route.

Proposition 3 (Rectangular kernel exclusion solves singleton). *Assume the colored centered extension theorem for colored centers with fewer than $|V(K)|$ vertices. Assume also the no nonzero rectangular kernel statement for the fixed-host singleton state (K, S, T, a, b) . Then the fixed-host singleton switch is solved.*

Proof sketch. The rectangular second-boundary identity holds for every equal-deck singleton state. By no nonzero rectangular kernel, at least one endpoint cancellation holds:

$$H_a \cong H_b \quad \text{or} \quad C_b^a \cong C_a^b.$$

In the first case, the direct-edge size-drop lemma applies. In the second case, the complementary direct-cancellation size-drop lemma applies. Either way, the problem recenters over a host with one fewer vertex, and the colored centered induction hypothesis gives the required color-preserving automorphism of K carrying $T \cup \{a\}$ to $T \cup \{b\}$. $\square$

So the fixed-host singleton problem can now be attacked without mentioning type-flow paths at all. Type flows remain useful for finding a minimal obstruction, but the rectangular kernel is the algebraic object that must be proved empty. This is a meaningful simplification: instead of showing that no routing through T and no routing through O can coexist, it is enough to show that the common $T \times O$ two-hole boundary separates one of the two endpoint differences.

5.20 The second-order boundary issue

There is a tempting but invalid shortcut. Each type-flow edge

$$B_t \rightarrow A_t$$

compares two singleton colorings of the smaller fixed host $K - t$:

$$B_t = (K - t, S \setminus \{t\}, (T \setminus \{t\}) \cup \{b\}),$$

$$A_t = (K - t, S \setminus \{t\}, (T \setminus \{t\}) \cup \{a\}).$$

If these two smaller colorings had equal fixed-host decks, the induction hypothesis would solve the edge. But original deck equality does not give that equality edge-by-edge. It gives only the aggregate balance of all such edges, plus the two endpoint shadows H_a, H_b .

Thus a proof cannot simply apply induction to every type-flow edge. The failure of a single edge to have equal smaller decks is a second-order boundary. When one deletes another vertex $w \neq t$, the discrepancy between $A_t - w$ and $B_t - w$ must be compensated elsewhere in the global second-deletion data.

This suggests a hierarchy. Let $\mathcal{G}_m$ be the free abelian group on isomorphism classes of finite colored graphs with m vertices, and define the deletion boundary

$$\partial[X] = \sum_{v \in V(X)} [X - v].$$

The fixed-host singleton equality says that, for the two colored full hosts

$$X_a = (K, S, T \cup \{a\}), \quad X_b = (K, S, T \cup \{b\}),$$

we have

$$\partial[X_a] = \partial[X_b].$$

Equivalently,

$$\partial([X_a] - [X_b]) = 0.$$

The direct-shadow theorem asks for one specific cancellation inside this boundary: the term obtained by deleting a on the left should match the term obtained by deleting b on the right.

Remark 16 (Why this is not a free lunch). *For arbitrary colored graphs, injectivity of ∂ is just the colored reconstruction problem. So the boundary language does not by itself prove anything. Its value here is that $[X_a] - [X_b]$ is a very special one-vertex color difference inside a fixed host. The hope is to prove acyclicity for this special class of chains, not for all graph chains at once.*

Conjecture 30 (Singleton boundary acyclicity). *Let $X_a = (K, S, T \cup \{a\})$ and $X_b = (K, S, T \cup \{b\})$ be a fixed-host singleton pair. If*

$$\partial([X_a] - [X_b]) = 0,$$

then the natural boundary terms cancel:

$$[K - a, S \setminus \{a\}, T] = [K - b, S \setminus \{b\}, T].$$

This is exactly the fixed-host direct-shadow theorem, rewritten as a special acyclicity statement. A shortest positive type-flow path is a candidate nontrivial cycle in this singleton boundary complex. The absorber exchange, witness-charge, and neutral-reversibility constraints are attempts to prove that such a special cycle cannot exist.

Conjecture 31 (Higher-order path shortening). *If a shortest positive singleton type-flow path exists, then examining its second-order boundary either produces a shorter positive type-flow path or a direct natural-shadow match $H_a \cong H_b$.*

This is the broader version of exchange-chain shortening. It says that second-deletion discrepancies cannot merely circulate at the next level; they must expose a shorter obstruction. If this higher-order shortening fails, then the approach has likely run into genuine homology of the colored deletion complex, which would explain why the current strategy cannot close the theorem without a new idea.

Conjecture 32 (No closed fixed-host singleton flow). *In the fixed-host singleton switch, the two natural shadow types*

$$H_a = (K - a, S \setminus \{a\}, T), \quad H_b = (K - b, S \setminus \{b\}, T)$$

cannot both be routed entirely through detours in T , avoiding the direct positions b and a , respectively.

The computational evidence suggests a still stronger and cleaner statement.

Conjecture 33 (Fixed-host direct-shadow theorem). *In every fixed-host singleton switch with equal colored deletion decks,*

$$(K - a, S \setminus \{a\}, T) \cong (K - b, S \setminus \{b\}, T).$$

Equivalently, the natural shadow of the moved vertex on one side always matches the natural shadow of the moved vertex on the other side.

The direct-shadow theorem immediately implies no closed singleton flow. It is stronger than what the induction needs, but it may be the right local lemma: the deck appears unable to route the natural singleton shadow through the common second color at all.

This is now the most local detour-selection target. A counterexample would not merely need deletion-similar vertices in T ; it would need a balanced population of absorbers in the same color and degree-profile layer, with all proper moving-vertex profiles still agreeing. The next invariant to try is a two-step count relative to T : vertices adjacent to exactly one of a, b should create signed shifts in these detour cards, just as the witness sets W^- and W^+ did in the original single-switch analysis.

5.21 Witness charges in the fixed-host singleton switch

Define the second-color two-step shadow

$$Q_z(U) = \sum_{x \in U \setminus \{z\}} \deg_{K-z}(x).$$

Equivalently, $Q_z(U)$ counts oriented length-two walks in the deleted colored host $K - z$ whose first vertex lies in the second color $U \setminus \{z\}$. This is a numerical invariant of the deleted two-colored card.

Lemma 35 (Fixed-host two-step defect). *In the singleton setting $U = T \cup \{a\}$, $V = T \cup \{b\}$, we have*

$$\begin{aligned} Q_a(U) - Q_a(V) &= -\deg_{K-a}(b), \\ Q_b(U) - Q_b(V) &= \deg_{K-b}(a), \end{aligned}$$

and for every $z \notin \{a, b\}$,

$$Q_z(U) - Q_z(V) = \mathbf{1}_{bz \in E(K)} - \mathbf{1}_{az \in E(K)}.$$

Proof sketch. For $z \notin \{a, b\}$, the common contribution of the vertices in $T \setminus \{z\}$ cancels. The remaining difference is

$$\deg_{K-z}(a) - \deg_{K-z}(b) = \deg_K(a) - \deg_K(b) - \mathbf{1}_{az \in E(K)} + \mathbf{1}_{bz \in E(K)}.$$

The low-order singleton balance gives $\deg_K(a) = \deg_K(b)$, leaving the displayed formula. The cases $z = a$ and $z = b$ are the same computation with one of the moving vertices absent from one side's second color. $\square$

Define the fixed-host witness sets

$$X^- = \{z \notin \{a, b\} : az \in E(K), bz \notin E(K)\},$$

$$X^+ = \{z \notin \{a, b\} : bz \in E(K), az \notin E(K)\}.$$

Then degree balance gives

$$|X^-| = |X^+|,$$

and the two-step defect formula says

$$Q_z(U) - Q_z(V) = \begin{cases} -1, & z \in X^-, \\ +1, & z \in X^+, \\ 0, & \text{otherwise} \end{cases}$$

for $z \notin \{a, b\}$.

Thus the fixed-host singleton switch has the same signed witness geometry as the original single-switch problem. If a and b are twins relative to the whole host, then the transposition $a \leftrightarrow b$ preserves K ; because $a \in S$ if and only if $b \in S$, and because both are outside T , it also preserves S and carries $T \cup \{a\}$ to $T \cup \{b\}$. So any counterexample must have nonempty balanced witness sets X^- and X^+ .

After choosing a fixed-host trade, the same telescoping argument as cycle balance applies to Q . On every cycle of the trade permutation, the aligned Q -defects must sum to zero. Hence witnesses cannot be hidden independently: every cycle that carries a positive witness charge must also carry a compensating negative charge, with the special vertices a and b contributing the larger endpoint charges

$$-\deg_{K-a}(b), \quad +\deg_{K-b}(a).$$

Conjecture 34 (No nonterminal fixed-host witness cycle). *In a fixed-host singleton counterexample, there is no trade cycle that balances the endpoint charges and the witness charges while routing both natural shadows through T and avoiding a zero-star-error direct edge.*

This is the fixed-host version of the edge-slide obstruction. The smallest non-twin case has

$$|X^-| = |X^+| = 1.$$

Then any nonterminal trade cycle has essentially one lost neighbor and one gained non-neighbor to account for. The expected contradiction is that deleting those witnesses exposes the slide unless the direct $a \rightsquigarrow b$ edge already has a zero-error representative.

5.22 Direct edges as a size drop

The direct-edge case has an important extra feature: even if the direct isomorphism does not have zero star error, it lowers the size of the centered problem.

Suppose we have a direct fixed-host card edge

$$\eta : (K - a, S \setminus \{a\}, T) \cong (K - b, S \setminus \{b\}, T).$$

Identify these two colored cards with a common colored host

$$(L, S_L, T_L).$$

The two full colored hosts

$$(K, S, T \cup \{a\}), \quad (K, S, T \cup \{b\})$$

are now two one-vertex extensions of L . The re-added vertex has the same first-color status on both sides and is second-colored on both sides. Its two attachment sets are

$$A_\eta = N_K(a) \setminus \{a\}, \quad B_\eta = \eta^{-1}(N_K(b) \setminus \{b\}),$$

viewed as subsets of $V(L)$. Their symmetric difference is exactly the direct star error of η .

Lemma 36 (Direct recentring drops the host size). *Under fixed-host singleton deck equality, a direct card edge produces a colored centered state over L , with*

$$|V(L)| = |V(K)| - 1,$$

and

$$\text{PDeck}_L(A_\eta) = \text{PDeck}_L(B_\eta)$$

in the colored sense, where the common colors S_L, T_L are part of the card data.

Proof sketch. The two colored full hosts have equal vertex decks by hypothesis. The direct card edge identifies one card on each side with the same colored host (L, S_L, T_L) . Cancelling this matched card from the two colored decks leaves equality of the punctured colored extension decks over L . The center has one fewer vertex because $L \cong K - a \cong K - b$. $\square$

Conjecture 35 (Colored centered extension theorem). *Fix a finite graph L together with any finite list of vertex colors $\mathcal{C}$. For attachment sets $A, B \subseteq V(L)$, form the one-vertex extensions in which the new vertex has the same prescribed color data on both sides. If the punctured decks of these colored extensions agree as colored cards, then the two colored full extensions are isomorphic.*

This is not meant as a fundamentally new conjecture. A finite list of colors can be encoded by adjoining a bounded amount of rooted gadget data, or treated directly in a Lean formalization as extra predicates on vertices. The point of naming it is bookkeeping: direct recentring from a fixed-host singleton switch lands in the same centered problem, but with passive colors carried along.

Remark 17 (Simultaneous colored induction). *The proof should not appeal to the colored centered theorem as an external black box. Instead, run the whole centered-extension induction simultaneously for finite colored centers. The induction statement at size n is:*

for every finite graph C with any fixed finite list of vertex colors, and for attachment sets $A, B \subseteq V(C)$, if the colored punctured extension decks agree and the added root has the same color data on both sides, then the two colored extensions are isomorphic.

The uncolored centered theorem is the special case with no colors. All moving-hole, star-error, cycle-balance, and detour arguments are unchanged: card isomorphisms are simply required to preserve the passive colors. The direct-edge size-drop step is then a legitimate appeal to the induction hypothesis at size $n - 1$, not a circular appeal to a stronger theorem at the same size.

Proposition 4 (Direct edge plus smaller colored centered theorem). *Assume the colored centered extension theorem is known for colored centers with fewer than $|V(K)|$ vertices. Then any fixed-host singleton switch with a direct card edge is solved: there is an automorphism of K preserving S and carrying $T \cup \{a\}$ to $T \cup \{b\}$.*

Proof sketch. Choose a direct card edge and form the smaller colored centered state over (L, S_L, T_L) . If the direct star error is empty, the extended direct isomorphism is already the desired automorphism. Otherwise the smaller colored centered theorem applies to the equal punctured decks over L , so the two one-vertex colored extensions over L are isomorphic. Transporting this isomorphism back through the direct recentring gives a color-preserving isomorphism

$$(K, S, T \cup \{a\}) \cong (K, S, T \cup \{b\}),$$

which is exactly an element of $\text{Aut}(K, S)$ carrying $T \cup \{a\}$ to $T \cup \{b\}$. $\square$

Thus the fixed-host singleton problem splits cleanly. Direct edges are inductive size-drop moves. The same-size obstruction is detour selection: show that the natural shadows cannot both be hidden inside T forever. In this sense, the closed-flow conjecture is the real fixed-host singleton heart; neighborhood selection after a direct edge can be delegated to the smaller centered theorem.

Proposition 5 (Inductive singleton package). *Assume the following two statements.*

1. *The colored centered extension theorem holds for colored centers with fewer than $|V(K)|$ vertices.*
2. *The no closed fixed-host singleton flow conjecture holds for the fixed background (K, S, T) .*

Then the fixed-host singleton switch over (K, S, T) is solved.

Proof sketch. If there is a direct card edge $a \rightsquigarrow b$, the direct-edge size-drop proposition applies and gives the desired color-preserving automorphism. Suppose no direct card edge exists. By natural shadow isolation, the natural a -shadow must be routed through some detour vertex of T . Applying the same argument with a and b interchanged, the natural b -shadow must also be routed through T . This is exactly the closed singleton flow forbidden by the second assumption. Hence the no-direct case is impossible, and the direct case has already been handled. $\square$

This is a useful strategic compression. We no longer need to prove fixed-host singleton rigidity in one breath. It is enough to prove the same-size detour-selection statement; all direct failures are passed down to a smaller centered problem. This is the first place where a genuine induction on the number of vertices looks structurally plausible.

Consequently, if no automorphism of the background (K, S, T) sends a to b , then any rooted separator between a and b must use the whole host. The singleton obstruction is therefore a full-size pseudo-similarity: all proper moving-vertex profiles agree, and the only possible distinction is the full colored orbit. This is precisely the boundary where the detour-flow and star-error arguments have to enter.

Proposition 6 (Fixed-host reconstruction implies neighborhood selection). *The equal-size fixed-host subset reconstruction conjecture implies the neighborhood selection conjecture.*

Proof sketch. After choosing a direct card isomorphism and transporting to

$$K = C - p,$$

neighborhood selection asks whether

$$U = N_C(p) \setminus \{p\} \quad \text{and} \quad V = \psi^{-1}(N_C(q) \setminus \{q\})$$

are in the same $\text{Aut}(K, S)$ -orbit. Direct recentering gives an anchored state with equal punctured decks. Once the distinguished first root is kept as part of the anchored data, the subdeck obtained by deleting ordinary vertices of K is precisely the fixed-host deletion deck, so we get

$$\text{Deck}_{K,S}(U) = \text{Deck}_{K,S}(V).$$

The two subsets have equal size because $\deg_C(p) = \deg_C(q)$. Fixed-host subset reconstruction therefore gives the desired orbit equality. $\square$

5.23 Pattern-sum invariants

Let P be a finite graph whose vertices carry two colors: the first color corresponds to membership in S , and the second color corresponds to membership in U . Define the pattern-sum invariant

$$\pi_P(U)$$

to be the number of induced subsets $R \subseteq V(K)$ such that the two-colored induced structure

$$(K[R], S \cap R, U \cap R)$$

is isomorphic to P .

The full list of pattern-sum invariants determines the $\text{Aut}(K, S)$ -orbit of U : in particular, the full-size pattern

$$(K, S, U)$$

separates U from any subset not in its orbit. The fixed-host deck and anchored Kelly pressure explain which of these invariants are already forced.

Lemma 37 (Proper pattern sums are deck-controlled). *Under the hypotheses of equal-size fixed-host deck equality,*

$$\text{Deck}_{K,S}(U) = \text{Deck}_{K,S}(V), \quad |U| = |V|,$$

we have

$$\pi_P(U) = \pi_P(V)$$

for every two-colored pattern P with fewer vertices than K .

Proof sketch. The invariant $\pi_P(U)$ is exactly a proper colored induced-pattern count in the fixed host. Anchored Kelly pressure reconstructs this count from the fixed-host deletion deck. Equal decks therefore give equal proper pattern sums. $\square$

Thus a fixed-host counterexample is exactly a pair of equal-size subsets U, V that agree on every proper pattern-sum invariant but disagree on their full orbit. This is an invariant-theoretic restatement of the full-size colored hypomorphism obstruction.

Conjecture 36 (No full-degree-only subset obstruction). *For every finite colored host (K, S) , equal-size subsets U, V that agree on all proper pattern-sum invariants are in the same $\text{Aut}(K, S)$ -orbit.*

This conjecture is equivalent in spirit to fixed-host subset reconstruction, but it removes the card language. It says that equal-size subsets of a fixed colored host have no orbit distinction visible only at full degree. If false, a counterexample would be a very explicit finite pattern-counting phenomenon: two nonconjugate subsets with identical proper colored induced substructure statistics.

5.24 Recursive view of neighborhood selection

Fix a direct card isomorphism

$$\psi : C - p \cong C - q, \quad \psi(S) = S,$$

and transport the q -side back to

$$K = C - p.$$

Inside K , define two neighborhood attachment sets

$$U = N_C(p) \setminus \{p\}, \quad V = \psi^{-1}(N_C(q) \setminus \{q\}).$$

The direct card isomorphism extends over $p \mapsto q$ exactly when

$$U = V.$$

If $U \neq V$, then the direct recentering produces a new centered state whose defect is

$$U \triangle V.$$

In fact the new center is

$$D = \text{Ext}(K, S),$$

and the two re-added vertices are attached to

$$U \cup \{*\} \quad \text{and} \quad V \cup \{*\},$$

where $*$ is the root of $\text{Ext}(K, S)$. The root belongs to both attachment sets, so the new defect is still $U \triangle V$.

Thus neighborhood selection is exactly the assertion that the direct recentered state is terminal. If it is not terminal, the star error

$$|U \triangle V|$$

is the new defect. In the single-switch base case, parity forces this new defect to be either 0 or at least 2. Exact two-error is precisely the case in which $U \triangle V = \{a, b\}$, giving the slide from the switched pair (p, q) to the witness pair (a, b) .

This recursive view explains the slide graph: it is the graph of repeated direct recenterings in the special situation where the defect remains 2. The zero-companion lemma says that this graph has no nonterminal vertices reachable from a genuine same-deck single switch.

5.25 Root-anchored centered states

The direct recentering produces a special kind of centered state. The new center is

$$D = \text{Ext}(K, S),$$

with distinguished root $*$, and both new attachment sets contain $*$:

$$A' = U \cup \{*\}, \quad B' = V \cup \{*\}.$$

Thus the defect avoids the distinguished root. Moreover, automorphisms of D fixing $*$ are exactly automorphisms of K preserving S :

$$\text{Aut}(D, *) \cong \text{Aut}(K, S).$$

So neighborhood selection is equivalent to saying that the root-anchored state (D, A', B') is terminal by an automorphism fixing the distinguished root.

Lemma 38 (Direct recentering preserves punctured deck equality). *In the single-switch setting, suppose a direct root-preserving card isomorphism identifies*

$$\text{Ext}(C - p, S) \cong \text{Ext}(C - q, S)$$

with a common center $D = \text{Ext}(K, S)$. Let

$$A' = U \cup \{*\}, \quad B' = V \cup \{*\}$$

be the two recentered attachment sets described above. Then

$$\text{PDeck}_D(A') = \text{PDeck}_D(B').$$

Proof sketch. The recentered presentations describe the same two full graphs as before:

$$\text{Ext}(C, S \cup \{p\}) \cong \text{Ext}(D, A'), \quad \text{Ext}(C, S \cup \{q\}) \cong \text{Ext}(D, B').$$

The original graphs have equal full decks, and the direct card match identifies one occurrence of D on each side. Cancelling that common card from the full decks gives equality of the punctured extension decks over D . $\square$

Conjecture 37 (Root-anchored centered theorem). *Let D be a centered graph with a distinguished vertex $*$. Suppose*

$$A' = \{*\} \cup U, \quad B' = \{*\} \cup V,$$

and

$$\text{PDeck}_D(A') = \text{PDeck}_D(B').$$

*Then either U and V are in the same orbit under $\text{Aut}(D, *)$, or there is a recentering move that keeps the distinguished root in both attachment sets and strictly decreases $|U \Delta V|$.*

If this anchored theorem holds, then neighborhood selection follows in the single-switch situation: the direct recentering gives a root-anchored state with equal punctured decks. If it is terminal, the stabilizer-orbit lemma selects an extendable direct card isomorphism. If it descends, then in the defect-2 base case the only possible strict descent is defect 0, again giving the desired zero-error companion.

This anchored version may be easier than the full centered theorem because the distinguished root is common to both attachment sets. It removes one source of ambiguity: the defect lives entirely among old vertices of K , while $*$ records the original color set S .

Proposition 7 (Detour selection plus anchored descent proves the base case). *Assume the following two statements.*

1. *In every single-switch state with equal punctured extension decks, the rooted card-match graph contains the direct edge $p \rightsquigarrow q$.*
2. *The root-anchored centered theorem holds for the direct recentered state.*

Then the single-switch rigidity conjecture holds.

Proof sketch. By the first assumption, choose a direct root-preserving card isomorphism

$$\psi : C - p \cong C - q, \quad \psi(S) = S.$$

Direct recentering gives the root-anchored state

$$(D, A', B') = (\text{Ext}(K, S), U \cup \{*\}, V \cup \{*\})$$

with

$$\text{PDeck}_D(A') = \text{PDeck}_D(B').$$

By the anchored theorem, either U and V are in the same $\text{Aut}(D, *)$ -orbit, or there is a strict anchored descent. Iterating the anchored theorem must terminate because the anchored defect is finite. At termination the two anchored full extensions are isomorphic, so the original single-switch full extensions are isomorphic as well. If termination happens already at the first anchored state, the stabilizer-orbit formulation gives an extendable direct card isomorphism, hence an automorphism of C carrying $S \cup \{p\}$ to $S \cup \{q\}$. $\square$

5.26 Two-root interpretation

The root-anchored state can also be read as a graph with two adjacent special vertices. Start with $K = C - p$. Add a first root r adjacent to S . Then add a second root x adjacent to r and to U . The resulting graph is

$$\text{Ext}(\text{Ext}(K, S), U \cup \{r\}).$$

The comparison state replaces U by V :

$$\text{Ext}(\text{Ext}(K, S), V \cup \{r\}).$$

Thus the neighborhood-selection problem is a two-color reconstruction problem on K : the first color S is encoded by the neighbors of r , and the second color U or V is encoded by the neighbors of x . The two roots are adjacent in both graphs.

In this language, an automorphism fixing r and carrying U to V is exactly an isomorphism of the two-root structures that fixes the first root and sends the second root to the second root. A failure of neighborhood selection would therefore be a pair of two-colorings (S, U) and (S, V) of the same graph K whose associated two-root graphs have the same deck, but whose second colors are not in the same $\text{Aut}(K, S)$ -orbit.

This is a useful warning as well as a simplification. General colored graph reconstruction is not obviously easier than graph reconstruction. The hope is that the present two-root situation is special: the first root is common, the second root is adjacent to it on both sides, and the state arose from a single-switch comparison, so the second-color defect begins in the tightly controlled neighborhood set $U \Delta V$.

5.27 Minimal bad form for the single-switch theorem

Putting the pieces together, a counterexample to the single-switch theorem would have to satisfy a rigid list of conditions.

1. $A = S \cup \{p\}$, $B = S \cup \{q\}$, and $\text{PDeck}_C(A) = \text{PDeck}_C(B)$.
2. p and q are degree-balanced:

$$\deg_C(p) = \deg_C(q).$$

3. p and q are not twins outside $\{p, q\}$, since otherwise twin collapse gives the automorphism.
4. The witness sets

$$W^- = \{w : pw \in E(C), qw \notin E(C)\}, \quad W^+ = \{w : qw \in E(C), pw \notin E(C)\}$$

have equal nonzero size.

5. The witnesses are balanced relative to the common color:

$$\deg_S(p) = \deg_S(q).$$

6. More generally, p and q have identical proper root-link profiles: every proper induced pattern using the added root occurs equally often in $\text{Ext}(C, S \cup \{p\})$ and $\text{Ext}(C, S \cup \{q\})$.
7. The rooted card-match graph admits perfect matchings, but no direct edge $p \rightsquigarrow q$ with zero star error.
8. If the natural shadow $\text{Ext}(C - p, S)$ is not isolated, every detour absorbing it must pass through vertices of S with the same degree and compatible rooted triangle/two-step profiles.
9. If a direct card edge $p \rightsquigarrow q$ exists, every corresponding S -preserving isomorphism $C - p \cong C - q$ must fail neighborhood selection, producing a nonempty even neighborhood defect.

The computation through six vertices says this normal form is empty. The proof campaign is now to show it is empty in general. The two plausible routes are:

detour selection/isolation or neighborhood selection.

Either one would prove the single-switch base case.

This is a smaller target than the full single-switch rigidity conjecture. It does not ask us to understand the whole deck at once; it only asks us to show that the deck equality forces one of the two defect-deleted shadows to match with no old-card neighborhood error.

5.28 Shape of an exact two-error

The first bad case to exclude is therefore very specific. Suppose a root-preserving match based at p gives a bijection $f = f_p$ with

$$f(A) = B$$

and

$$|\text{Err}_p(f)| = 2.$$

Let $v = f(p)$. Since the two compared neighborhoods have the same size, there are unique vertices $a, b \in V(C) \setminus \{p\}$ such that, after transport by f ,

$$a \in N_C(p), \quad f(a) \notin N_C(v),$$

and

$$b \notin N_C(p), \quad f(b) \in N_C(v).$$

Every other adjacency from p is preserved by f , and every adjacency not involving p is preserved by f . Thus f is an isomorphism from C to the graph obtained from C by deleting the edge pa and

adding the edge pb , after relabelling by f . In other words, the obstruction is a single edge-slide at the deleted vertex.

This suggests a more concrete base-case attack. A single-switch counterexample with no zero-error match would have to make every defect-deleted card look correct while hiding an edge-slide at the missing vertex. But the two witnesses a and b reappear as actual deleted vertices in their own shadows. Deleting a removes the lost-edge witness; deleting b removes the gained-edge witness. The deck should have to match these two shadows in a way that is compatible with the same color transport, and that compatibility is where the contradiction ought to live.

Conjecture 38 (Edge-slide exposure). *In the single-switch setting, an exact two-error match at p forces a distinguishable imbalance between the shadows obtained by deleting the lost neighbor a and the gained non-neighbor b , unless a zero-error match already exists.*

Conjecture 39 (Zero companion for edge-slides). *In the single-switch setting, suppose a defect-rooted root-preserving card match has exact star error 2. Then there is another defect-rooted root-preserving card match with star error 0.*

This is the sharpened base-case target suggested by computation. Exact two-error matches are not themselves contradictory; they appear even in small examples where the switched colorings are globally isomorphic. The claim is that an exact edge-slide cannot be isolated. If the deck permits the hidden edge pa to slide to pb , then the other shadows must also contain a companion alignment that repairs the slide completely.

Remark 18 (Fallback: edge-slide descent). *The zero-error target may be stronger than necessary. If a defect-2 single-switch state admits an exact two-error recentering, then recentering does not decrease the raw defect; the new defect again has size 2. But it changes the switched pair. In the notation above, the new switched vertices are precisely the lost/gained witnesses a and b inside the punctured card. Thus an exact two-error move is not noise: it moves the single switch from (p, q) to the local edge-slide witness pair (a, b) .*

This suggests a secondary base-case potential. For a switched pair p, q , define

$$\omega_C(p, q) = \frac{1}{2} |\{w \notin \{p, q\} : (pw \in E(C)) \nleftrightarrow (qw \in E(C))\}|,$$

so the comparison ignores the possible edge pq . The twin case is $\omega_C(p, q) = 0$. The smallest non-twin case is $\omega_C(p, q) = 1$, exactly one lost witness and one gained witness. A weaker but perhaps more realistic base-case theorem would be: every exact two-error recentering either gives an isomorphism or moves to a new single-switch state with smaller value of some refinement of ω , for example the multiset of rooted profiles of the witness pair. Then defect 2 would terminate by this secondary potential even without an immediate zero-error match.

5.29 The slide graph at defect two

The fallback can be organized as a finite directed graph. A vertex of the slide graph is a defect-2 centered state

$$(C, S, p, q), \quad A = S \cup \{p\}, \quad B = S \cup \{q\},$$

with $\text{PDeck}_C(A) = \text{PDeck}_C(B)$. A terminal vertex is one admitting a zero-error defect-rooted match. A directed edge is an exact two-error recenter:

$$(C, S, p, q) \longrightarrow (D, S', a, b),$$

where a is the lost neighbor and b is the gained non-neighbor in the edge-slide description above. Here D is the matched punctured card used as the new center, and the new attachment sets have the form

$$A' = S' \cup \{a\}, \quad B' = S' \cup \{b\}.$$

The single-switch base case follows from either of the following statements.

Conjecture 40 (No nonterminal slide cycle). *Every vertex in the defect-2 slide graph reaches a terminal vertex.*

Conjecture 41 (Slide potential). *There is a finite-ranking potential $\Omega(C, S, p, q)$, defined on defect-2 states, such that every nonterminal exact two-error recentering strictly decreases Ω after choosing the recentering appropriately.*

The witness sets give the first candidate for Ω . For a switched pair p, q , let

$$W^- = \{w : pw \in E(C), qw \notin E(C)\}, \quad W^+ = \{w : qw \in E(C), pw \notin E(C)\},$$

again outside $\{p, q\}$. The quantity

$$\omega_C(p, q) = |W^-| = |W^+|$$

is the coarsest measure of non-twinness. If $\omega = 0$, we are terminal by twin collapse. If $\omega = 1$, every exact two-error slide uses the unique pair of witnesses. If $\omega > 1$, one can hope to choose a witness pair whose recentering lowers ω , or at least lowers the signed multiset of rooted witness profiles.

This reframes the base case as a small dynamical problem: a counterexample is not just a pair of switched vertices. It is a closed slide system in which the deck keeps moving the switch from one witness pair to another without ever reaching twins or a zero-error card match.

The most promising invariants for exposing the edge-slide are rooted two-step and triangle counts. In a shadow $\text{Ext}(C - w, A \setminus \{w\})$, the number of triangles using the root is exactly

$$|E(C[A \setminus \{w\}])|.$$

Similarly, the number of rooted length-two paths from the root through the card records

$$\sum_{x \in A \setminus \{w\}} \deg_{C-w}(x).$$

For $A = S \cup \{p\}$ and $B = S \cup \{q\}$, these quantities change in controlled ways when w is one of p, q, a, b . The base-case proof should try to show that an edge-slide cannot preserve all of these rooted local profiles across the full punctured deck.

Here are the aligned formulas. Write

$$\deg_R(x) = |N_C(x) \cap R|$$

for the number of neighbors of x in a set $R \subseteq V(C)$. Also write

$$T_w(A) = |E(C[A \setminus \{w\}])|$$

for the number of root-using triangles in the w -deleted shadow, and

$$P_w(A) = \sum_{x \in A \setminus \{w\}} \deg_{C-w}(x)$$

for the number of oriented root-using two-step walks whose first step leaves the root. In the single-switch case $A = S \cup \{p\}$ and $B = S \cup \{q\}$, the aligned triangle differences are

$$T_p(A) - T_p(B) = -\deg_S(q),$$

$$T_q(A) - T_q(B) = \deg_S(p),$$

and, for $w \notin \{p, q\}$,

$$T_w(A) - T_w(B) = \deg_{S \setminus \{w\}}(p) - \deg_{S \setminus \{w\}}(q).$$

The aligned two-step differences are

$$P_p(A) - P_p(B) = -\deg_{C-p}(q),$$

$$P_q(A) - P_q(B) = \deg_{C-q}(p),$$

and, for $w \notin \{p, q\}$,

$$P_w(A) - P_w(B) = \deg_{C-w}(p) - \deg_{C-w}(q).$$

Using degree balance, this last formula becomes

$$P_w(A) - P_w(B) = \mathbf{1}_{qw \in E(C)} - \mathbf{1}_{pw \in E(C)}.$$

Thus every vertex w that distinguishes p from q produces a visible ± 1 shift in the aligned rooted two-step count.

Define the witness sets

$$W^- = \{w \notin \{p, q\} : pw \in E(C), qw \notin E(C)\}, \quad W^+ = \{w \notin \{p, q\} : qw \in E(C), pw \notin E(C)\}.$$

Equivalently,

$$W^- = (N_C(p) \setminus N_C(q)) \setminus \{q\}, \quad W^+ = (N_C(q) \setminus N_C(p)) \setminus \{p\}.$$

Degree balance gives

$$|W^-| = |W^+|.$$

For $w \notin \{p, q\}$, the two-step formula says

$$P_w(A) - P_w(B) = \begin{cases} -1, & w \in W^-, \\ +1, & w \in W^+, \\ 0, & \text{otherwise.} \end{cases}$$

So a non-twin single-switch counterexample must hide a balanced collection of negative and positive witness shadows. The smallest possible obstruction has

$$|W^-| = |W^+| = 1,$$

which is exactly the edge-slide picture above.

After choosing a centered trade, the cycle-balance lemma below makes this stronger. For the rooted two-step invariant P , every cycle of the trade permutation must contain balanced P -defect. Ignoring the special vertices p and q , that means a cycle cannot contain a W^+ -witness without also containing compensating W^- -defect, and conversely. Thus the deck cannot hide witnesses independently; it must organize them into balanced cycles of the matching permutation.

More explicitly, let

$$r = \deg_{C-p}(q) = \deg_{C-q}(p).$$

Then the aligned P -defect is the signed charge distribution

$$\delta_P(p) = -r, \quad \delta_P(q) = +r,$$

$$\delta_P(w) = -1 \quad (w \in W^-), \quad \delta_P(w) = +1 \quad (w \in W^+),$$

and $\delta_P(w) = 0$ for all remaining vertices. Every cycle of any centered trade must have total charge zero. Thus a minimal single-switch counterexample is not just balanced globally; its witness charges must be partitioned into zero-sum cycles compatible with the full rooted card types. For $\omega_C(p, q) = 1$, this leaves very few combinatorial possibilities.

Deck equality says that the multiset of the full rooted profiles

$$\{(T_w(A), P_w(A), \dots) : w \in V(C)\}$$

equals the corresponding B -side multiset. The aligned formulas show what has to be hidden by the matching. In particular, if p and q have different adjacency profiles into S , the rooted triangle shadows already force a nontrivial rematching pattern. If all these first moments agree, then p and q are already very similar relative to S , which is exactly the situation where a zero-error automorphism becomes more plausible.

Lemma 39 (Twin collapse). *If p and q have the same neighbors outside $\{p, q\}$, then the transposition swapping p and q and fixing all other vertices is an automorphism of C . Consequently it carries*

$$S \cup \{p\} \quad \text{to} \quad S \cup \{q\},$$

so the single-switch case is solved.

Proof sketch. All edges not incident to p or q are fixed. For any $x \notin \{p, q\}$, the edge px exists if and only if qx exists, so the transposition preserves the edges from the pair $\{p, q\}$ to the rest of the graph. The edge pq , if present, is fixed as an unordered edge. $\square$

Thus a single-switch counterexample must be non-twin: there is some witness vertex x adjacent to exactly one of p, q . The low rooted moments above measure such witnesses. The remaining difficulty is that deck equality may permute the shadows, so the proof has to show that the witness cannot be hidden globally by a rematching of all cards.

So the one-step descent conjecture has a concrete form: if the identity comparison between A and B has color error

$$A \triangle B,$$

then some card match based at a defect vertex should give a color-preserving bijection with $\hat{\psi}(A) = B$ whose one-star graph error is smaller than that color error. If the one-star error is empty, $\hat{\psi}$ is already the desired automorphism of C .

Conjecture 42 (One-step defect descent). *Suppose*

$$\text{PDeck}_C(A) = \text{PDeck}_C(B)$$

and A and B are not in the same $\text{Aut}(C)$ -orbit. Then there exist

$$u \in A \triangle B, \quad v \in V(C),$$

and a root-preserving isomorphism

$$\theta : \text{Ext}(C - u, A \setminus \{u\}) \cong \text{Ext}(C - v, B \setminus \{v\})$$

such that the recentered defect satisfies

$$|\Delta_\theta(u, v)| < |A \triangle B|.$$

Proposition 8 (Descent proves the centered theorem). *The one-step defect descent conjecture implies the centered extension theorem, and hence the reconstruction conjecture.*

Proof sketch. Assume the centered extension theorem fails, and choose a counterexample (C, A, B) with $|A \triangle B|$ minimal, so

$$\text{PDeck}_C(A) = \text{PDeck}_C(B) \quad \text{but} \quad \text{Ext}(C, A) \not\cong \text{Ext}(C, B).$$

The defect is nonzero. In particular A and B are not in the same $\text{Aut}(C)$ -orbit, so one-step descent applies. Recenter through a root-preserving matched card to obtain a new centered presentation

$$(D, A', B')$$

of the same two full extension graphs with

$$|A' \triangle B'| < |A \triangle B|.$$

Because the full extension decks agree and the recentering identifies the chosen root-deleted cards, cancellation gives

$$\text{PDeck}_D(A') = \text{PDeck}_D(B').$$

By minimality, A' and B' lie in the same $\text{Aut}(D)$ -orbit, so the two full extensions are isomorphic. Transporting back through the recentering shows $\text{Ext}(C, A) \cong \text{Ext}(C, B)$, contradicting that (C, A, B) was a counterexample. The centered reduction then gives reconstruction. $\square$

Remark 19. *This proposition proves the centered extension theorem, not automatically the stronger root-preserving centered theorem. After recentering, the original root is just a vertex of the new center D ; a terminal automorphism of (D, A', B') may move it. To prove the stronger theorem, one would need to carry an additional marked-root invariant through the state graph. For the reconstruction conjecture itself, no such extra marking is needed.*

5.30 The star-error campaign

The proof campaign is now sharply focused. Suppose a positive-defect counterexample exists with

$$d = |A \triangle B|$$

minimal. A root-preserving centered trade supplies, for each deleted vertex that participates in the trade, a bijection

$$f_u : V(C) \rightarrow V(C)$$

with three properties:

$$f_u(A) = B,$$

f_u preserves all adjacencies away from the star of u ,

and

$$|\text{Err}_u(f_u)| = |\Delta(u, f_u(u))|.$$

Minimality says that no available f_u , with $u \in A\Delta B$, has

$$0 < |\text{Err}_u(f_u)| < d.$$

Thus every useful one-star error is either zero, which proves the theorem, or has size at least d . This is an extremely rigid normal form.

The next idea is to compare two such near-automorphisms. If f_u and f_w are two root-preserving trade maps, set

$$\gamma_{u,w} = f_w^{-1} f_u.$$

Then

$$\gamma_{u,w}(A) = A.$$

Moreover, $\gamma_{u,w}$ can fail to preserve the graph structure only near two stars:

$$\text{star}(u) \quad \text{and} \quad \text{star}(f_u^{-1}(f_w(w))).$$

Indeed, f_u only creates errors at u , and f_w^{-1} only creates errors at the preimage of the f_w -hole.

Conjecture 43 (Two-hole rigidity). *In a minimal positive-defect centered trade, the transition maps $\gamma_{u,w}$ cannot all be nontrivial two-star errors. More precisely, some comparison of two trade maps either produces an automorphism preserving A , or produces a one-star recentering move whose error is strictly smaller than $d = |A\Delta B|$.*

This cannot be a statement about arbitrary two-star errors. Any transposition of two same-colored vertices preserves the color set A and can only change edges incident to those two vertices. Such maps are everywhere. The useful rigidity has to use the whole deck-indexed family of one-star maps, together with the fact that each f_u comes from an actual rooted card isomorphism. So the object to rule out is not an isolated two-star near-automorphism, but a closed coherent system of them arising from a centered trade.

This is the most plausible route to proving one-step descent. The deck gives many one-star near-automorphisms from (C, A) to (C, B) . A genuine counterexample would force all of them to have large star error, but their pairwise transition maps are color-preserving near-automorphisms of (C, A) with error supported on at most two stars. The hope is that such a large family of low-support graph errors cannot close up unless it comes from actual automorphisms.

5.31 An averaging target

There is a slightly softer version of one-step descent that may be easier to prove. For a chosen root-preserving trade, write f_u for the extended one-star map at u , and define the defect-star energy

$$\mathcal{E}_D(f) = \sum_{u \in A\Delta B} |\text{Err}_u(f_u)|.$$

If one can choose the trade so that

$$\mathcal{E}_D(f) < |A\Delta B|^2,$$

then some defect vertex u satisfies

$$|\text{Err}_u(f_u)| < |A\Delta B|,$$

which is exactly one-step descent.

Thus a minimal counterexample must satisfy the opposite inequality for every available root-preserving trade:

$$\mathcal{E}_D(f) \geq |A \triangle B|^2.$$

This is a useful target because $\mathcal{E}_D$ is an honest sum of local adjacency errors. It may be possible to compare it with the color transport cost of the same trade. Since each f_u carries A to B , the identity color defect $A \triangle B$ is being paid for by moving colored vertices; the conjectural inequality says that the deck cannot force every such payment to reappear as at least the same amount of graph error at every defect vertex.

5.32 Case 2: the root does not map to the root

If θ sends the root of the left extension to a non-root vertex on the right, then the shadow already exhibits root ambiguity. This should be treated as a potential direct success: it may allow the root of $\text{Ext}(C, A)$ to be identified with an old vertex of $\text{Ext}(C, B)$, producing an isomorphism of the full extensions after one more extension step.

Conjecture 44 (Root-moving lemma). *If some punctured-card isomorphism between $\text{Ext}(C - u, A \setminus \{u\})$ and $\text{Ext}(C - v, B \setminus \{v\})$ moves root to non-root, then either $\text{Ext}(C, A) \cong \text{Ext}(C, B)$, or there is a centered presentation with strictly smaller defect.*

Conjecture 45 (Root-preserving extraction). *In a minimal centered counterexample, after applying all possible root-moving success or ambiguity-descent moves, there is a centered trade whose card isomorphisms are root-preserving.*

This conjecture is the gateway into the charge-cycle machinery. The intended proof structure is:

$$\text{root-moving match} \Rightarrow \text{success or ambiguity descent},$$

while

$$\text{no useful root-moving match} \Rightarrow \text{root-preserving trade}.$$

Once the latter trade is chosen, the color-boundary and cycle-balance lemmas apply.

6 Root recognition and ambiguity

The root-moving case deserves its own invariant. In a centered extension $\text{Ext}(C, A)$, the root is certainly a vertex whose deletion gives C . But it need not be the only such vertex. Define the root-ambiguous set

$$R_C(A) = \{u \in V(C) : \text{Ext}(C - u, A \setminus \{u\}) \cong C\}.$$

These are the old vertices whose deletion produces a card isomorphic to the root-deleted card. Equivalently, they are the old vertices that can masquerade as the root from the point of view of the deck.

A first necessary condition is obtained from edge counts. If $u \in R_C(A)$, then

$$|E(C)| = |E(C - u)| + |A| - \mathbf{1}_{u \in A},$$

so

$$\deg_C(u) + \mathbf{1}_{u \in A} = |A|.$$

Thus a root-ambiguous old vertex has the same degree in the full extension as the root. Higher card invariants impose stronger versions of this condition.

If $R_C(A) = \emptyset$, then the root-deleted card C is unique in $\text{Deck}(\text{Ext}(C, A))$. In that case, every deck isomorphism must match root card to root card, and all remaining work is root-preserving.

If $R_C(A) \neq \emptyset$, then choosing $u \in R_C(A)$ and recentering at the card $\text{Ext}(C, A) - u \cong C$ produces a new centered presentation of the same full graph in which an old vertex has become the root. This is not a failure mode; it is a symmetry/ambiguity of the extension. It suggests a second ranking parameter:

$$\rho(C, A, B) = |R_C(A)| + |R_C(B)|.$$

Conjecture 46 (Root-ambiguity descent). *If a root-moving punctured-card isomorphism does not already extend to an isomorphism $\text{Ext}(C, A) \cong \text{Ext}(C, B)$, then recentering through it decreases either the attachment defect or the root-ambiguity parameter ρ .*

This would give a lexicographic potential of the form

$$(|A \triangle B|, \rho(C, A, B), \text{higher card-profile data}).$$

The root-preserving case changes attachment defect into old-card neighborhood defect; the root-moving case should consume root ambiguity.

7 State graph and potential functions

It may be too optimistic to expect the raw defect size $|A \triangle B|$ to decrease after every useful recentering. The root-preserving formula shows why: after recentering at u and v , the new defect is not directly the old attachment defect; it is the mismatch between the old-card neighborhoods of u and v :

$$(N_C(u) \setminus \{u\}) \triangle \psi^{-1}(N_C(v) \setminus \{v\}).$$

So a defect in the attachment set can be traded for a defect in the card structure.

This suggests replacing the one-shot descent lemma by a finite state graph. A state is a centered presentation

$$(C, A, B)$$

of the same two full graphs G and H . A move chooses an old vertex $u \in V(C)$, matches the punctured card $\text{Ext}(C - u, A \setminus \{u\})$ to some punctured card $\text{Ext}(C - v, B \setminus \{v\})$, and recenters through a chosen isomorphism.

The centered theorem is equivalent to saying that every state with $\text{PDeck}_C(A) = \text{PDeck}_C(B)$ can reach a terminal state:

$$A = B \quad \text{or, more generally,} \quad A \text{ and } B \text{ are in the same } \text{Aut}(C)\text{-orbit.}$$

Conjecture 47 (No positive-defect sink). *In the recentering state graph, every nonterminal state has at least one outgoing move to a state with smaller value for some finite-ranking potential $\Phi(C, A, B)$.*

Candidate potentials, in increasing order of sophistication, are:

$$\Phi_0(C, A, B) = |A \triangle B|,$$

$$\Phi_1(C, A, B) = (|A \triangle B|, \{\{\deg_C(w) : w \in A \triangle B\}\}),$$

with lexicographic order, and a more structural potential

$$\Phi_2(C, A, B) = \{ \{ (C - w, A \setminus \{w\}, B \setminus \{w\}) : w \in A \triangle B \} \},$$

ordered by the multiset extension of a recursively defined card-complexity order.

The advantage of the state-graph view is that a proof need not show raw defect always decreases. It is enough to rule out a directed cycle of genuinely positive-defect states in which every recentering merely moves the defect around. Such a positive cycle would be a “balanced trade” of attachment sets and old-card neighborhoods. The proof should aim to show that balanced trades cannot exist.

8 Normal form for a hypothetical counterexample

Assume the centered extension theorem is false, and choose a counterexample (C, A, B) minimizing the eventual potential Φ . Thus $\text{PDeck}_C(A) = \text{PDeck}_C(B)$, but $\text{Ext}(C, A) \not\cong \text{Ext}(C, B)$. Even before Φ is finalized, a minimal counterexample should satisfy the following constraints.

1. A and B are not in the same $\text{Aut}(C)$ -orbit.
2. The defect $A \triangle B$ is nonempty, and $|A| = |B|$.
3. For every $u \in V(C)$, every matched punctured-card isomorphism

$$\text{Ext}(C - u, A \setminus \{u\}) \cong \text{Ext}(C - v, B \setminus \{v\})$$

either fails to produce a smaller state, or it already preserves the putative minimal potential.

4. In particular, if such an isomorphism is root-preserving, with restriction $\psi : C - u \cong C - v$, then the recentered defect

$$(N_C(u) \setminus \{u\}) \triangle \psi^{-1}(N_C(v) \setminus \{v\})$$

is nonempty; otherwise the empty-defect extension lemma gives a full isomorphism.

5. If a root-moving isomorphism exists, then root ambiguity is not consumed by recentering. Equivalently, root ambiguity must persist around a directed cycle of centered presentations.

Thus a counterexample cannot merely have an attachment defect. It must have a self-sustaining trade in which attachment defects, old-card neighborhood defects, and root ambiguities continually transform into one another without ever vanishing. The next proof task is to show that such a self-sustaining trade is impossible.

9 Trades

It is useful to name the object that a counterexample would have to supply.

Definition 9 (Rooted card-match graph). *Given C, A, B , define the rooted card-match graph*

$$\mathcal{M}(C, A, B)$$

to be the bipartite graph with left and right vertex sets both $V(C)$, where there is an edge $u \rightsquigarrow v$ if there exists a root-preserving isomorphism

$$\text{Ext}(C - u, A \setminus \{u\}) \cong \text{Ext}(C - v, B \setminus \{v\}).$$

Each edge $u \rightsquigarrow v$ carries a set of possible star-error values, one for each such isomorphism.

A root-preserving centered trade is exactly a perfect matching in $\mathcal{M}(C, A, B)$, together with a choice of one isomorphism for every matched edge. In this language, one-step descent asks for a defect-left vertex $u \in A \triangle B$ and an incident edge $u \rightsquigarrow v$ carrying a star error smaller than $|A \triangle B|$. The cycle-balance lemmas apply after choosing a perfect matching, but the existence of a good recentering is an edge-existence statement.

Definition 10 (Centered trade). *A centered trade from (C, A) to (C, B) consists of a bijection $\tau : V(C) \rightarrow V(C)$ and, for every $u \in V(C)$, an isomorphism*

$$\theta_u : \text{Ext}(C - u, A \setminus \{u\}) \cong \text{Ext}(C - \tau(u), B \setminus \{\tau(u)\}).$$

It is root-preserving if every θ_u sends root to root.

Equality $\text{PDeck}_C(A) = \text{PDeck}_C(B)$ is exactly the assertion that at least one centered trade exists. The stronger root-preserving centered theorem would say that the existence of such a trade already forces a global automorphism of C , after possibly choosing a better trade:

$$\exists \alpha \in \text{Aut}(C), \quad \alpha(A) = B.$$

For a root-preserving trade, each θ_u restricts to a colored-card isomorphism

$$(C - u, A \setminus \{u\}) \cong (C - \tau(u), B \setminus \{\tau(u)\}).$$

Thus a root-preserving trade is precisely a hypomorphism between the two one-colored structures (C, A) and (C, B) . The remaining question is whether every such hypomorphism is induced by an actual colored isomorphism.

Lemma 40 (Root-preserving trades preserve the deleted color). *If (τ, θ_u) is a root-preserving centered trade, then*

$$\tau(A) = B.$$

Proof sketch. The card isomorphism at u identifies the rooted degrees, so

$$|A| - \mathbf{1}_{u \in A} = |B| - \mathbf{1}_{\tau(u) \in B}.$$

The full extensions have equal edge count, hence $|A| = |B|$. Therefore

$$u \in A \iff \tau(u) \in B.$$

Since τ is a bijection, it carries A onto B . □

Lemma 41 (Cycle balance for rooted invariants). *Let I be any numerical invariant of rooted cards, and write*

$$I_A(u) = I(\text{Ext}(C - u, A \setminus \{u\})), \quad I_B(u) = I(\text{Ext}(C - u, B \setminus \{u\})).$$

Suppose (τ, θ_u) is a centered trade. Define the aligned defect

$$\delta_I(u) = I_A(u) - I_B(u).$$

Then on every cycle Γ of the permutation τ ,

$$\sum_{u \in \Gamma} \delta_I(u) = 0.$$

Proof sketch. The trade gives

$$I_A(u) = I_B(\tau(u)).$$

Thus

$$\delta_I(\tau(u)) = I_A(\tau(u)) - I_B(\tau(u)) = I_A(\tau(u)) - I_A(u).$$

Summing this identity around a τ -cycle telescopes. $\square$

This lemma is small but important. Equality of decks gives only a multiset equality until we choose a trade. Once a trade is chosen, every scalar rooted card invariant becomes a discrete potential whose aligned defects must balance on each cycle of the trade permutation. In the single-switch case, using rooted two-step counts, this forces the $+1$ and -1 witness signals to be paired cycle-by-cycle, not merely in aggregate.

Corollary 3 (Defect vertices lie in balanced trade cycles). *For the edge-count invariant,*

$$\delta_E(u) = \mathbf{1}_{u \in B} - \mathbf{1}_{u \in A}.$$

Hence every cycle of a centered trade contains equally many vertices of $A \setminus B$ and $B \setminus A$. In particular, in the single-switch case $A = S \cup \{p\}$, $B = S \cup \{q\}$, the vertices p and q lie in the same cycle of the trade permutation.

Proof sketch. The formula for δ_E is exactly the aligned edge-count defect lemma. Apply cycle balance to the edge-count invariant. In the single-switch case there is only one negative charge and one positive charge, so they must occur in the same cycle. $\square$

Lemma 42 (Color-boundary form of a root-preserving trade cycle). *Assume the centered trade is root-preserving. Let*

$$x_0, x_1, \dots, x_{m-1}$$

be a cycle of τ , indexed so that $\tau(x_i) = x_{i+1}$. Put

$$b_i = \mathbf{1}_{x_i \in B}.$$

Then

$$\mathbf{1}_{x_i \in A} = b_{i+1},$$

and hence

$$x_i \in B \setminus A \iff b_i = 1, b_{i+1} = 0,$$

while

$$x_i \in A \setminus B \iff b_i = 0, b_{i+1} = 1.$$

Thus the defect vertices on a root-preserving trade cycle are exactly the boundaries between B -colored and B -uncolored arcs, and the two defect signs alternate around the cycle.

Proof sketch. The root-preserving trade gives $\tau(A) = B$, equivalently

$$x_i \in A \iff \tau(x_i) = x_{i+1} \in B.$$

The displayed boundary descriptions follow immediately. $\square$

9.1 Defect cycles

The preceding lemma suggests a more economical normal form for a minimal counterexample. Once a root-preserving trade is chosen, each trade cycle is a cyclic word in two colors: vertices in B and vertices outside B . The defect vertices are exactly the color-boundaries of this cyclic word. Thus the total defect splits into disjoint even blocks, one block for each trade cycle that changes color.

Definition 11 (Defect cycle). *A defect cycle of a root-preserving trade is a cycle of τ containing at least one vertex of $A\triangle B$.*

Conjecture 48 (Atomic trade reduction). *If a centered counterexample exists, then there is one with a root-preserving centered trade having a single defect cycle.*

This is not yet proved, but it is a promising reduction target. If a trade has several defect cycles, the edge-count charge already balances on each cycle separately. One hopes to either recenter using a vertex on one cycle to obtain descent, or modify the trade to reduce the number of defect cycles. In an atomic counterexample, all positive and negative defect vertices lie on one alternating trade cycle; the problem is then to show that at least one boundary vertex on that cycle has small star error.

Conjecture 49 (Cycle-local descent). *Let Γ be a defect cycle, and let*

$$d_\Gamma = |\Gamma \cap (A\triangle B)|.$$

Unless the full extensions are already isomorphic, some boundary vertex $u \in \Gamma \cap (A\triangle B)$ admits a root-preserving recentering whose star error is strictly smaller than d_Γ .

Cycle-local descent would imply one-step defect descent immediately, since $d_\Gamma \leq |A\triangle B|$. It also specializes to the single-switch base case: then the unique defect cycle has two boundary vertices, so strict cycle-local descent again means zero star error.

Conjecture 50 (No nontrivial centered trade). *There is no centered trade $(C, A) \rightsquigarrow (C, B)$ with A and B in distinct $\text{Aut}(C)$ -orbits that is minimal under recentering.*

This is the clean obstruction. The moving-hole machinery is an attempt to take a nontrivial trade and manufacture a smaller one by replacing the center with one of the traded cards.

10 Minimal-defect proof sketch

Assume for contradiction that the reconstruction conjecture fails. Choose a counterexample G, H with the same deck and $G \not\cong H$. Among all choices of matched cards $G - x \cong H - y$, and all identifications of the matched card with a graph C , choose a centered presentation

$$G \cong \text{Ext}(C, A), \quad H \cong \text{Ext}(C, B)$$

with $|A\triangle B|$ minimal.

The defect is nonzero, because zero defect gives a root-preserving isomorphism. The needed local theorem is not that an arbitrary defect vertex works, but that some defect vertex admits a good match. By the one-step defect descent lemma, choose

$$u \in A\triangle B, \quad v \in V(C),$$

and a matched punctured-card isomorphism

$$\text{Ext}(C - u, A \setminus \{u\}) \cong \text{Ext}(C - v, B \setminus \{v\})$$

whose recentering improves the chosen potential.

Apply the moving-hole analysis to this good isomorphism.

- If the isomorphism moves root to root, the star-error formula says that the new defect is exactly the one-star graph error of the associated bijection f_u , with $f_u(A) = B$. The descent choice makes this smaller than the original attachment defect, contradicting minimality.
- If the isomorphism moves root to non-root, the root-moving success lemma either gives a full isomorphism $G \cong H$, contradiction, or decreases the secondary root-ambiguity part of the potential.

Therefore no positive-defect counterexample exists. Hence every same-deck pair is isomorphic.

11 Ehrenfeucht-Fraisse interpretation

The same idea can be read as an Ehrenfeucht-Fraisse strategy. A card alignment is a partial isomorphism with one hidden hole. Spoiler tries to pick a vertex in the defect set; Duplicator responds by moving the hole to a new matched card before the defect becomes fatal. A defect-minimizing repair lemma would imply that Duplicator can maintain coherence through all n rounds.

However, the direct centered-extension formulation seems cleaner for proof work. For finite graphs, an n -round EF win is essentially equivalent to isomorphism, so the EF game does not reduce the difficulty; it organizes it. The core mathematical claim remains the centered trade theorem.

12 Lean-facing milestones

The following milestones are ordered so that failed attempts still produce useful formal infrastructure.

1. Define $\text{Ext}(C, A)$ in Lean as a graph on $V(C) \oplus \mathbf{1}$.
2. Define $\text{PDeck}_C(A)$ as the multiset of isomorphism classes of $\text{Ext}(C - u, A \setminus \{u\})$.
3. Prove the centered reduction: same deck plus one matched card gives equality of punctured extension decks.
4. Define defect $A \triangle B$, positive defect $A \setminus B$, and negative defect $B \setminus A$.
5. Prove the aligned edge-count defect lemma.
6. Generalize edge-count shadows to induced-subgraph-count shadows for copies using the root.
7. Formalize moving-hole recentering: from an isomorphism of punctured extensions, construct the new centered presentation.
8. Prove the star-error lemma: a root-preserving punctured-card isomorphism extends to a color-preserving bijection whose only graph errors are incident to the deleted vertex.

9. Prove the parity lemma for star error.
10. Formalize the single-switch base case infrastructure: degree balance, rooted triangle counts T_w , rooted two-step counts P_w , witness sets W^- and W^+ , and twin collapse.
11. Prove single-switch root-link profile balance by subtracting root-avoiding subgraph counts in the common card C from deck-reconstructed total subgraph counts.
12. Formalize absorbers, absorber constraints, and the natural shadow isolation criterion.
13. Formalize the rooted card-match graph and the direct-edge selection problem.
14. Formalize neighborhood selection as an $\text{Aut}(C - p, S)$ -orbit problem.
15. Define the fixed-host subset deck $\text{Deck}_{K,S}(U)$, fixed-host trades, and the fixed-host star-error lemma: every matched deleted colored card extends to a color-preserving bijection whose graph error is concentrated in one star.
16. Prove the fixed-host rehosting lemma, converting a fixed-host orbit failure into a centered attachment-defect problem over a smaller colored card.
17. Formalize the fixed-host singleton switch: moving-vertex profile balance, low-order singleton balances, absorber constraints, and the second-color two-step witness charge formula.
18. Prove the direct-edge size-drop lemma for the fixed-host singleton switch, and package it with no closed singleton flow into an induction step.
19. Formalize the two-slice type-flow decomposition and the rectangular second-boundary identity
$$\partial_O([H_b] - [H_a]) = \partial_T([C_a^b] - [C_b^a]).$$
20. Attack the no nonzero rectangular kernel lemma. This is now the most concrete fixed-host singleton target: a proof gives the direct cancellation needed for the size-drop induction, while a counterexample gives a precise obstruction to the current campaign.
21. Prove the two-hole splice lemma, or find a locked two-hole match. This is the local step inside the rectangular kernel argument: every matched two-hole card should either give a smaller obstruction, a direct cancellation, or a chord shortening one of the two type-flow paths.
22. Define the rectangular matching potential Φ , choose a Φ -minimal rectangular matching, and prove the minimum-error rectangular matching lemma: no essential locked match survives after all partner swaps that reduce $(\Phi, \#\text{locks})$.
23. Define the root-anchored centered subproblem and prove that direct recentering preserves punctured deck equality.
24. Prove cycle balance for arbitrary rooted numerical invariants, with edge-count balance and color-boundary trade cycles as the first corollaries.
25. Attack the single-switch theorem via detour selection/isolation plus neighborhood selection.
26. Formalize transition maps between two star-error bijections, and prove the two-hole error-support bound.
27. Attack one-step defect descent, with two-hole rigidity as the main local obstruction.
28. Attack the root-moving success lemma as the secondary ambiguity case.

13 Current gap

The proof currently hinges on four increasingly local claims.

1. Root-moving matches either give an isomorphism or decrease a root-ambiguity potential, allowing us to extract a root-preserving trade in a minimal counterexample.
2. In a root-preserving trade, defect lies on color-boundaries of trade cycles. The desired local theorem is cycle-local descent: some boundary vertex on a defect cycle has star error smaller than the number of defect vertices on that cycle.
3. The base case of cycle-local descent is the single-switch case. The target has split into direct-edge selection and neighborhood selection.
4. In the hard single-switch case, all proper root-link profiles agree. A remaining counterexample must therefore be a full-size orbit failure in the root-anchored two-color state. The fixed-host trade analysis sharpens this: such a failure is a coherent family of color-preserving one-star near-automorphisms with no zero-error member.

The next English task is to attack this sharpened fixed-host obstruction. The most local target is the fixed-host singleton switch: prove that two pseudo-similar colored vertices relative to a fixed background (K, S, T) must actually be related by an automorphism. This has now split into two smaller claims: no closed fixed-host singleton flow through T , and no nonterminal fixed-host witness cycle balancing the signed two-step charges. Direct singleton edges are now size-drop moves: after direct recentering, the remaining neighborhood-selection failure is a smaller colored centered problem. The same-size detour obstruction has been sharpened further: prove that the singleton type-flow graph has no positive-length path from H_a to H_b . The complementary type-flow slice gives a second path through the non- T vertices; two-slice incompatibility is now another possible way to force shortening. The exchange-chain, neutral-reversibility, and witness-tail claims are proposed tools for shortening such paths. The higher-order boundary formulation has now been sharpened to a rectangular kernel problem:

$$\partial_O([H_b] - [H_a]) = \partial_T([C_a^b] - [C_b^a]).$$

If the rectangular boundary forces one endpoint difference to vanish, the fixed-host singleton theorem follows by size-drop induction. If not, the approach has found its likely break point: a genuine nonzero rectangular kernel for the colored deletion operator, rather than merely a missing low-order invariant. The rectangular kernel has itself been localized to the two-hole splice lemma, and then to the minimum-error rectangular matching lemma. A productive two-hole match gives a direct cancellation, a smaller fixed-host singleton state, or a chord in one of the two shortest type-flow paths. Thus the immediate make-or-break target is the essential locked case: a two-hole match whose four ports stay internal to T and O , whose low and high extension errors are nonzero or color-incompatible, and which survives in a rectangular matching minimizing

$$(\Phi, \# \text{essential locks}).$$